SEA.AI

SEA.AI WATCHKEEPER

INSTALLATION MANUAL



SEA.AI Professional Services

SEA.AI
Port La Forêt
29940 La Forêt Fouesnant,
France

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Precaution of use

For safe and reliable operation of the SEA.AI system, please observe the following precautions. Failure to follow these instructions may result in fire, electric shock, personal injury, loss of vessel control, or voiding of the warranty.

Electrical Safety

⚠ WARNING – Risk of Electric Shock

- Disconnect power before installation or servicing. Always switch off the vessel's power supply before beginning installation or maintenance. Failure to do so may result in fire, electric shock, or serious injury.

⚠ WARNING – Fire Hazard

- Verify correct power supply. Ensure that the vessel's power source matches the rated voltage of the unit. Connection to an incorrect power supply may cause fire, equipment damage, or hazardous operation.

Installation Safety

⚠ CAUTION – Qualified Personnel Only

- The SEA.AI system must be installed and set up by a qualified marine electronics technician, in accordance with the installation instructions. Improper installation, modification, or use not in accordance with these instructions may compromise safety, affect performance, and void the warranty.

⚠ WARNING – Do Not Disassemble

- The unit must not be opened, altered, or modified by unauthorized personnel. Doing so may cause fire, electric shock, or serious injury, and will void the warranty.

⚠ NOTICE – Camera Alignment

- Ensure correct alignment of the masthead vision unit. The adjustment and orientation of the camera unit have a significant impact on system performance. Improper alignment may reduce detection accuracy and compromise the effectiveness of the system.

Operational Safety

⚠ NOTICE – Navigation Responsibility

- Do not rely solely on the SEA.AI system. For the safety of the vessel and all persons on board, the captain must always maintain a proper lookout and use all available means of navigation and collision-avoidance in accordance with the International Regulations for Preventing Collisions at Sea (COLREGs). SEA.AI is intended as an aid to navigation and does not replace the responsibility of the captain or crew.

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Limited Warranty

SEA.AI GmbH provides a limited warranty covering manufacturing defects in materials and workmanship for a period of two (2) years from the date of purchase, under normal use and service conditions.

This limited warranty gives you specific legal rights. You may also have other rights that vary by country or jurisdiction. SEA.AI GmbH does not exclude, limit, or suspend any rights you may have under applicable consumer protection laws. For a full understanding of your rights, please consult the laws of your country.

The complete terms, conditions, limitations, and exclusions of this limited warranty are set out in the SEA.AI Terms & Conditions.

Declaration of Conformity

The full text of the EU Declaration of Conformity is available upon request.

Product Documentation

The complete product manuals, including installation, operation, and safety instructions for the SEA.AI system, are available for download at Watchkeeper Document Library.

Software Update

You may need to update the SEA.AI Watchkeeper software after installation to ensure optimal performance and the latest features.

Important: Follow the software update instructions carefully. Updates should be performed according to the procedures in this manual.

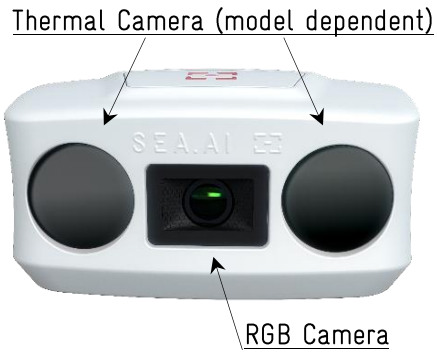
For detailed instructions, please refer to section 6. Watchkeeper Software Update of this manual.

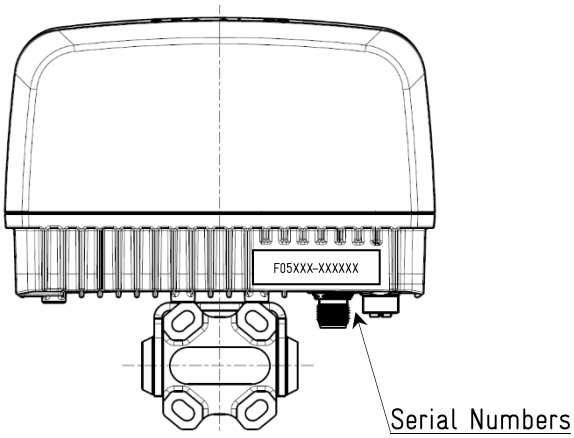
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1 INTRODUCTION

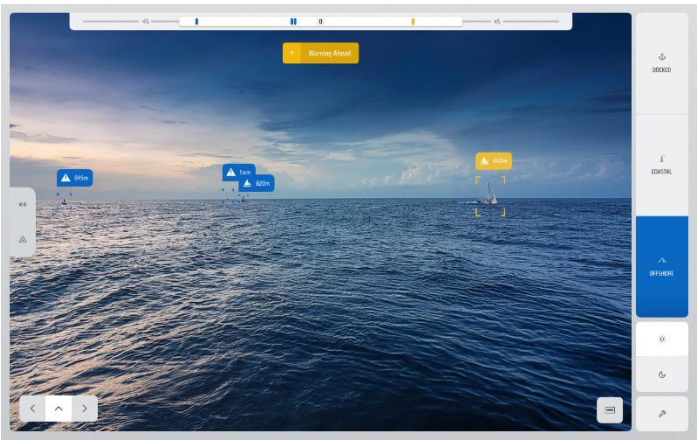
1.1 System Overview and Specifications

- The **SEA.AI Watchkeeper** uses optical sensors and Artificial Intelligence to detect objects along your vessel's trajectory. It automatically assesses the risk of collision in real time and issues alerts if a potential hazard is detected.
- Installing a SEA.AI system on board enhances the safety of your vessel, crew, and passengers.
- All SEA.AI systems can be controlled via the **SEA.AI Navigation App**, which is compatible with smartphones and tablets (Android and iOS), chartplotters, and computers.

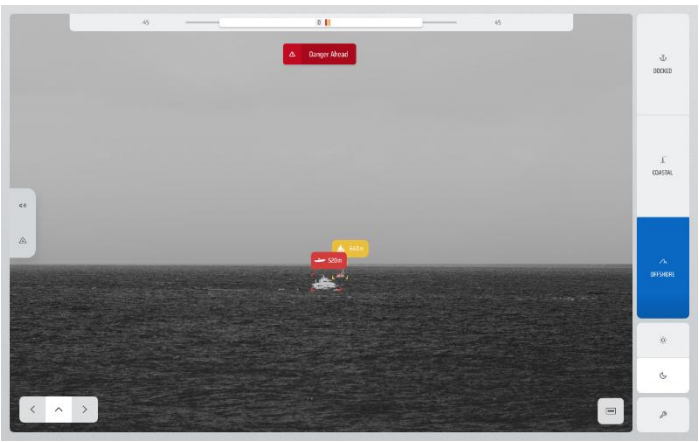




User Interface, RGB Camera View



User Interface, Thermal Camera View



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1.2 Motor Yacht System Specification

Watchkeeper for Powerboats	Lite	320	640	1280
				
	Affordable solution with lowlight camera and an ultra-wide field of view.	Additional 320x256px Long Wave Thermal Camera providing night vision even in full darkness.	High resolution thermal camera (640x512) for higher detection range and early warning. Increases the time to react.	2x 640 Thermal Cams =1280 horizontal resolution resulting in wide field of view without compromising detection range.
IR Sensor	-	1 Long Wave Infrared Thermal Camera	1 Long Wave Infrared Thermal Camera	2 Long Wave Infrared Thermal Camera
Resolution IR (thermal)	-	320x256	640x512	1280x512
FoV Thermal (HxV)	-	24°x19° (9.1mm)	24°x19° (18mm)	48°x19° (2x18mm)
Detection Range Thermal* *(buoy/RIB/Yacht)*	-	110m / 500m / 1300m	220m / 960m / 2500m	220m / 960m / 2500m
Color Sensor	4k low-light with HDR	4k low-light with HDR	4k low-light with HDR	4k low-light with HDR
Resolution Color	3840x2160	3840x2160	3840x2160	3840x2160
FoV Color (H/V)	130° x 65°	130° x 65°	130° x 65°	130° x 65°
Detection Range low-light** *(buoy/RIB/Yacht)*	65m / 350m / 900m	65m / 350m / 900m	65m / 350m / 900m	65m / 350m / 900m
Weight (Camera unit)	935g	951g	*975g*	*1023g*
Size	110x163x86mm	110x163x86mm	110x163x86mm	110x163x86mm
Environment and CE Marking	According IEC 60945	According IEC 60945	According IEC 60945	According IEC 60945
Waterproof rating	IP67	IP67	IP67	IP67
Connectivity	Ethernet, GPS, LTE	Ethernet, GPS, LTE	Ethernet, GPS, LTE	Ethernet, GPS, LTE
Electrical, Power supply	12-36V DC , 26.1W max	12-36V DC , 27.1W max	12-36V DC , 27.7W max	12-36V DC , 29.3W max

**Detection range varies depending on weather conditions e.g. sea state, rain, fog, humidity.*
***Detection range varies depending on light and weather conditions e.g. daytime, sea state, rain, fog, humidity.*

1.3 Motor Yacht Installation Kit (Art N°F055.. 10/20) Parts Included

Art N°	Description	Qty
F05XXX	Watchkeeper Unit	1
A04280	Mast Bracket	1
A04290	Mast Bracket Socket	1
A04410	Rosetta M8	2
A04420	Countersunk Taping Screw DIN7991 M8x25 A4	2
A04430	Countersunk Taping Screw DIN7991 M4x10 A4	4
A04800	Galvanic current isolation paste	1
A04460/A04140	Data Cable 10m/20m	1
A04470/A04150	Power Cable 10m/20m	1
Document	Watchkeeper User Guide	1

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1.4 Sailing Yacht System Specification

Watchkeeper S for Sailing Boats	**Lite S**	**320 S**	**640 S**	**1024 S**
				
	Affordable solution with lowlight camera and an ultra-wide field of view.	Additional 320x256px Long Wave Thermal Camera providing night vision even in full darkness.	High resolution thermal camera (640x512) resulting in wide field of view without compromising detection range.	2x 512x640 Thermal Cam =1024 horizontal resolution resulting in superior detection range without compromising detection angel.
IR Sensor	-	1 Long Wave Infrared Thermal Camera	1 Long Wave Infrared Thermal Camera	2 Long Wave Infrared Thermal Camera
Resolution IR (thermal)	-	320x256	640x512 t	1024x640
FoV Thermal (HxV)	-	34°x25° (6.3mm)	50°x40° (8.7 mm)	50°x34° (2x14mm)
Detection Range Thermal* *(buoy/RIB/Yacht)*	-	80m / 350m / 900m	100m / 460m / 1200m	170m / 750m / 2000m
Color Sensor	4k low-light with HDR	4k low-light with HDR	4k low-light with HDR	4k low-light with HDR
Resolution Color Sensor	3840x2160	3840x2160	3840x2160	3840x2160
FoV (H/V)	130° x 65°	130° x 65°	130° x 65°	130° x 65°
Detection Range low-light** *(buoy/RIB/Yacht)*	65m / 350m / 900m	65m / 350m / 900m	65m / 350m / 900m	65m / 350m / 900m
Weight (Camera unit)	935g	951g	*975g*	*1023g*
Size	110x163x86mm	110x163x86mm	110x163x86mm	110x163x86mm
Environment and CE Marking	According IEC 60945	According IEC 60945	According IEC 60945	According IEC 60945
Waterproof rating	IP67	IP67	IP67	IP67
Connectivity	Ethernet, GPS, LTE	Ethernet, GPS, LTE	Ethernet, GPS, LTE	Ethernet, GPS, LTE
Electrical, Power supply	12-36V DC , 26.1W max	12-36V DC , 27.1W max	12-36V DC , 27.7W max	12-36V DC , 29.3W max
Minimum Power Supply Voltage depending on cable length	>20-70m: 24V DC >70-100m: 36V DC	>20-70m: 24V DC >70-100m: 36V DC	>20-70m: 24V DC >70-100m: 36V DC	>20-70m: 24V DC >70-100m: 36V DC

**Detection range varies depending on weather conditions e.g. sea state, rain, fog, humidity. **Detection range varies depending on light and weather conditions e.g. daytime, sea state, rain, fog, humidity.*

1.5 Sailing Yacht Installation Kit (Art N°F055.. 30/40/50/60/70) Parts Included

Art N°	**Description**	**Qty**
F05XXX	Watchkeeper Sailing Unit	1
A04280	Mast Bracket	1
A04290	Mast Bracket Socket	1
A04300	Wind Sensor Bracket	1
A04410	Rosetta M8	2
A04420	Countersunk Taping Screw DIN7991 M8x25 A4	2
A04430	Countersunk Taping Screw DIN7991 M4x10 A4	4
A04800	Galvanic current isolation paste	1
A045XX	Data Cable 30m/35m/45m/55m/70m	1
A045XX	Power Cable 30m/35m/45m/55m/70m	1
Document	Watchkeeper User Guide	1

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1.6 Tools and Equipment Needed

- Bracket's Mounting screws (x4) – Refer to Chapter 3.8 for more details
- Power drill with appropriate drill bit
- Pen
- Tape Measure or ruler
- 5mm Allen key
- Necessary for cables routing
- Computer
- On-board network adapter

2 PRODUCT REGISTRATION, ACTIVATION, AND WARRANTY

To activate your SEA.AI Watchkeeper system and receive your **Product Key** for software updates, please register your product.

You can easily register by logging into your **SEA.AI Connect** account:

- **Login:** https://connect.sea.ai/account/login.
- **Create an account:** https://connect.sea.ai/account/register/step/1.

Product registration ensures your system remains up to date with the latest software enhancements and artificial intelligence improvements, maintaining optimal performance and safety.

Important: The product warranty period begins upon successful registration of your SEA.AI Watchkeeper system in SEA.AI Connect. Registration is therefore required to activate your warranty and enable access to software updates.

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3 INSTALLATION

3.1 General Installation Overview – Motor Yacht



3.2 Mounting Guidelines Motor and Sailing Yachts

Optimal Mounting Position:

- Select a location with the **most unobstructed field of view**. Objects such as boat structures, antennas, radars, or flags may block the camera's view and reduce system performance.

Elevation:

- Mount the Watchkeeper **as high as possible** on a stable and rigid support. Higher installations improve object detection range and distance prediction accuracy.

Thermal Detection:

- Avoid placing the Watchkeeper where **hot surfaces** (e.g., exhaust pipes) are in its direct field of view, as this can interfere with thermal detection.

Vibration Considerations:

- Choose a mounting location **free from excessive vibrations**, such as those generated by engines, to maintain stable performance.

Power supply:

- Install the unit **close to the vessel's DC power source** to minimize cable lengths and voltage drops.

Cables & connectors:

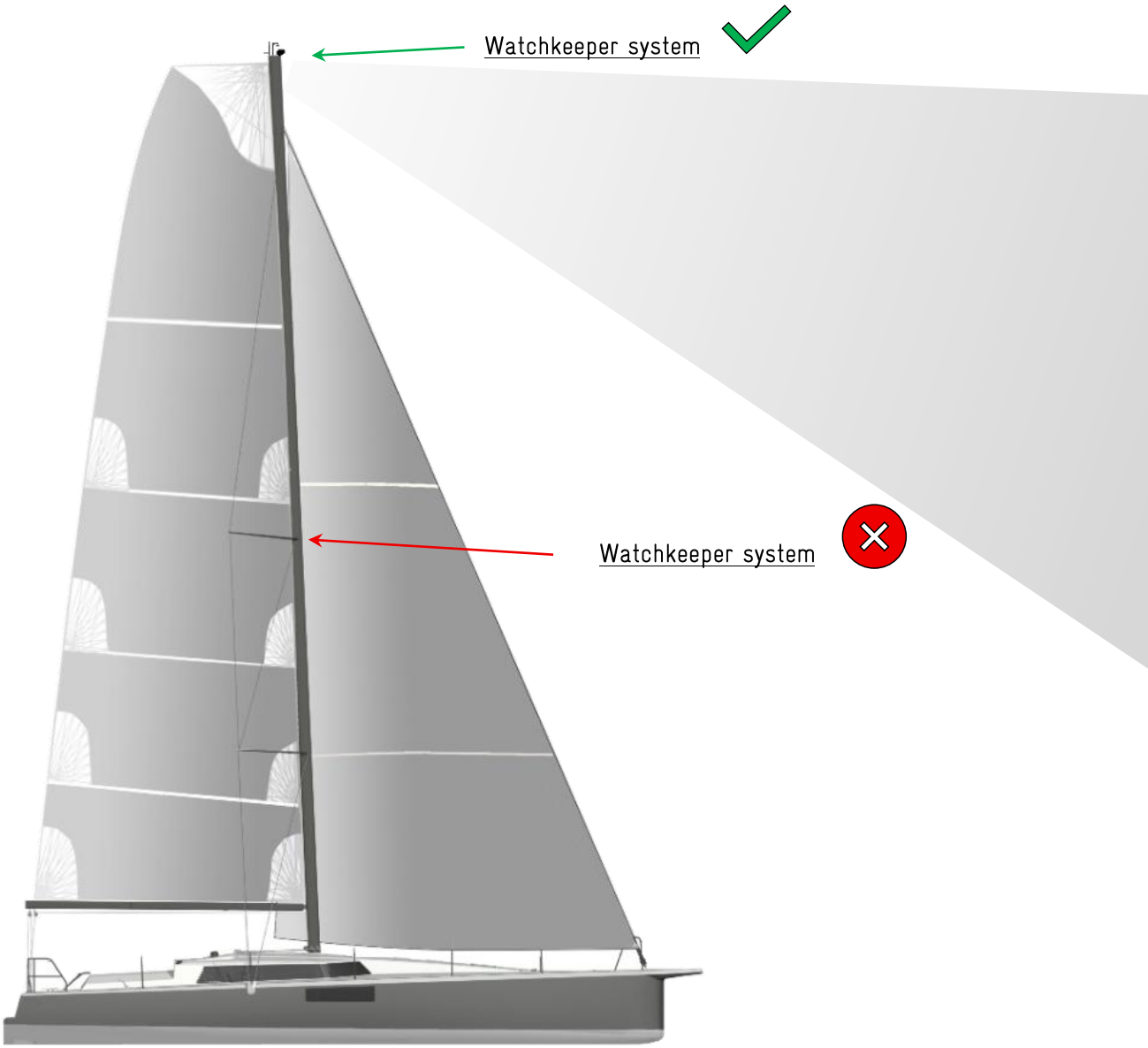
- Ensure there is **sufficient space behind the unit** for proper handling and secure connection of cables and electrical connectors.

Electrical Interference:

- Position the unit **away from devices that generate electrical interference**, such as radio transmitters or receivers.

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3.3 General Installation Overview – Sailing Yacht



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3.4 EMC Guidelines

The **SEA.AI Watchkeeper** complies with applicable Electromagnetic Compatibility (EMC) regulations, minimizing both electromagnetic emissions and susceptibility. **Proper installation is essential** to maintain optimal EMC performance.

Recommended Installation Guidelines for Best EMC Performance

- Maintain a minimum distance of **1 meter (3.28 feet)** from any radiant element of an S-band (or lower frequency) radar.
- Ensure at least **2 meters (6.56 feet)** of clearance from the path of a radar beam. Radar beams are assumed to spread **20 degrees above and below** the radiant element.
- Keep at least **1 meter (3.28 feet)** away from any equipment transmitting or cables carrying radio frequency signals, such as VHF radios, antenna, and related cabling.

3.5 Wiring

The Watchkeeper system has two harnesses; a power harness and data harness that connect to the back of the unit.

3.6 Wiring Guidelines

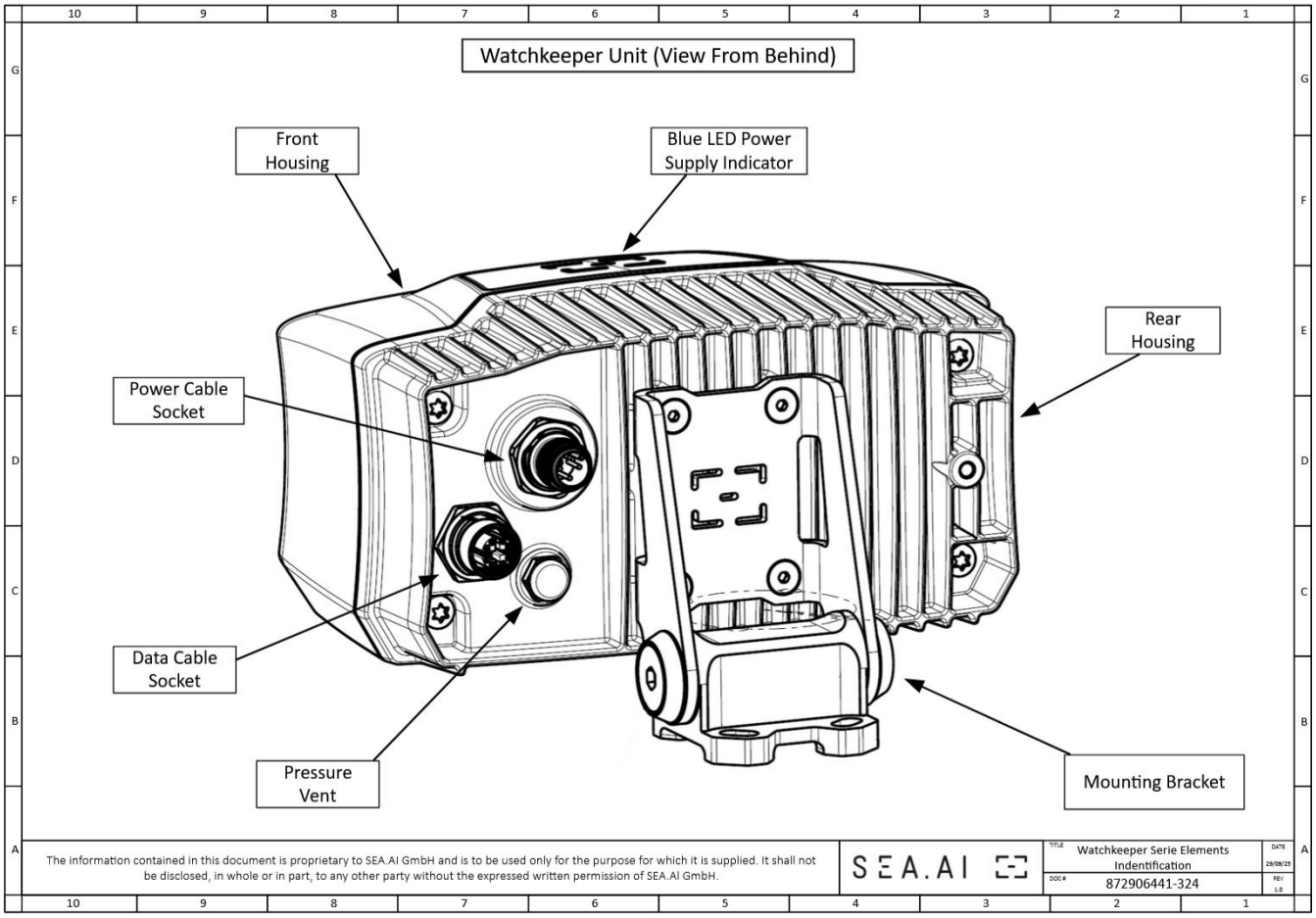
- Avoid kinking or over-bending the cables, and do not route cables over sharp edges or burrs.
- Use adequate strain relief to prevent excessive tension on the cables and connectors.
- Install cables in a manner that prevents water from flowing to the connectors.
- Protect the cables from any mechanical interference.
- Secure the cables properly and provide service loops and drip loops.
- Ensure there is enough space around the connectors for easy connection and disconnection of cables.
- Use a dedicated circuit breaker and fuse for Watchkeeper.
- Ensure all electrical connections are properly sealed and waterproof to prevent corrosion and short circuits.
- Use marine-grade cables and connectors that are rated for harsh maritime environment.
- Route cables away from high heat sources and moving parts to prevent damage.
- Regularly inspect cables and connectors for wear, corrosion, and secure connections, especially in high vibration areas.
- Use color-coded and labeled cables for easy identification and troubleshooting.
- Follow all relevant maritime regulations and standards for electrical installations.

⚠ **WARNING – Incorrect Polarity**

Connecting the power supply with incorrect polarity can permanently damage the SEA.AI Watchkeeper system. **Always ensure the power is switched off before beginning installation.**

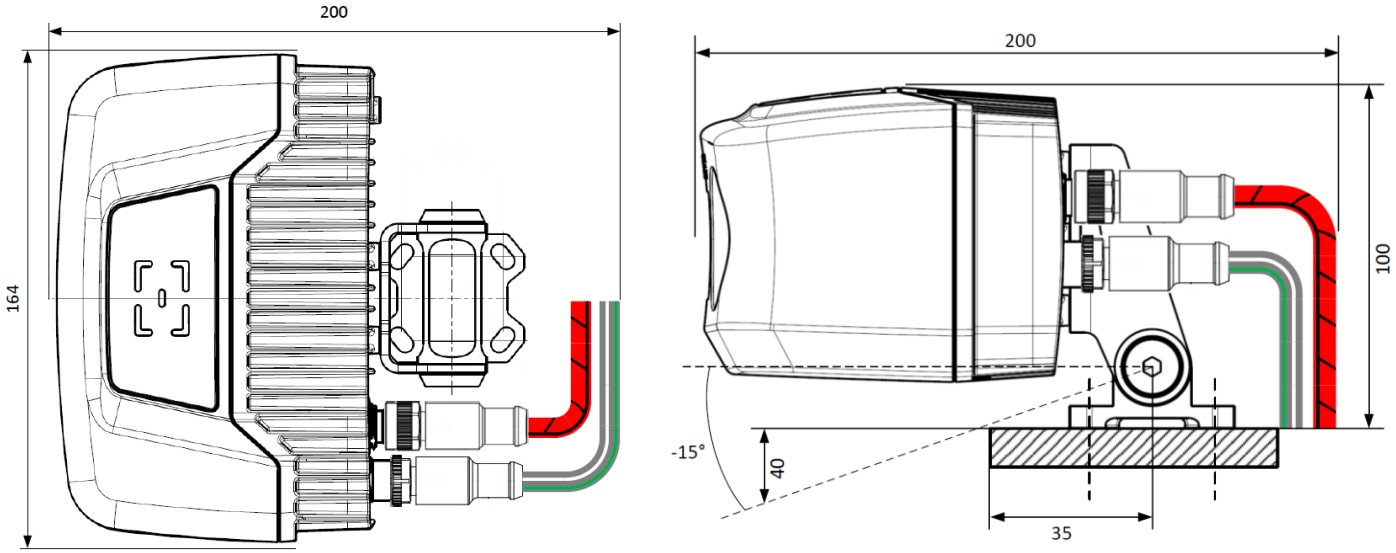
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3.7 Watchkeeper Overview



3.8 Mounting Clearance (in mm)

When installing the system, ensure that the dimensions of the unit, bracket, and cables allow free movement during vertical calibration and prevent any strain on the connectors.

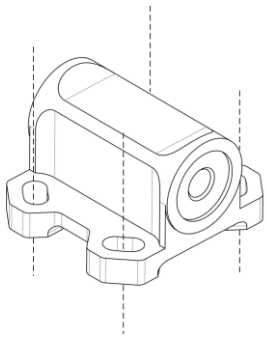


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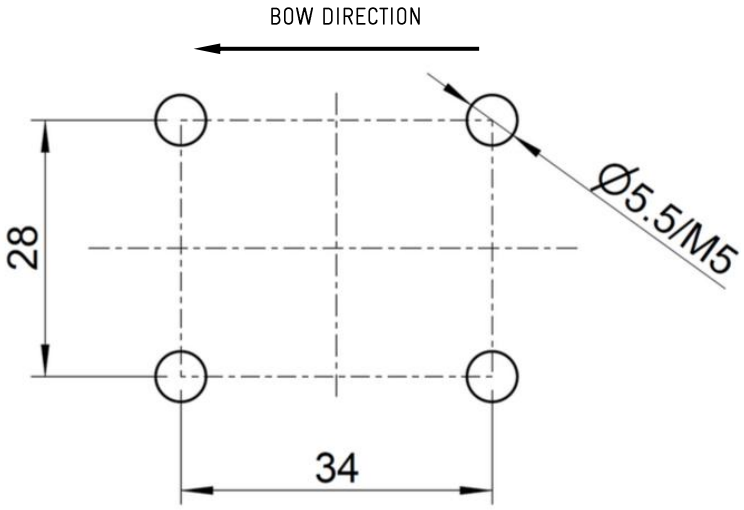
3.9 Mounting the Watchkeeper Bracket (A4280)

The following points must be observed when mounting the system:

- The system bracket must be secured using four 5 mm fasteners (not included).
- The mounting method must be adapted to the characteristics of the surface or structure on which the bracket is installed.
- The fasteners must be locked using a suitable thread locker or locknuts.



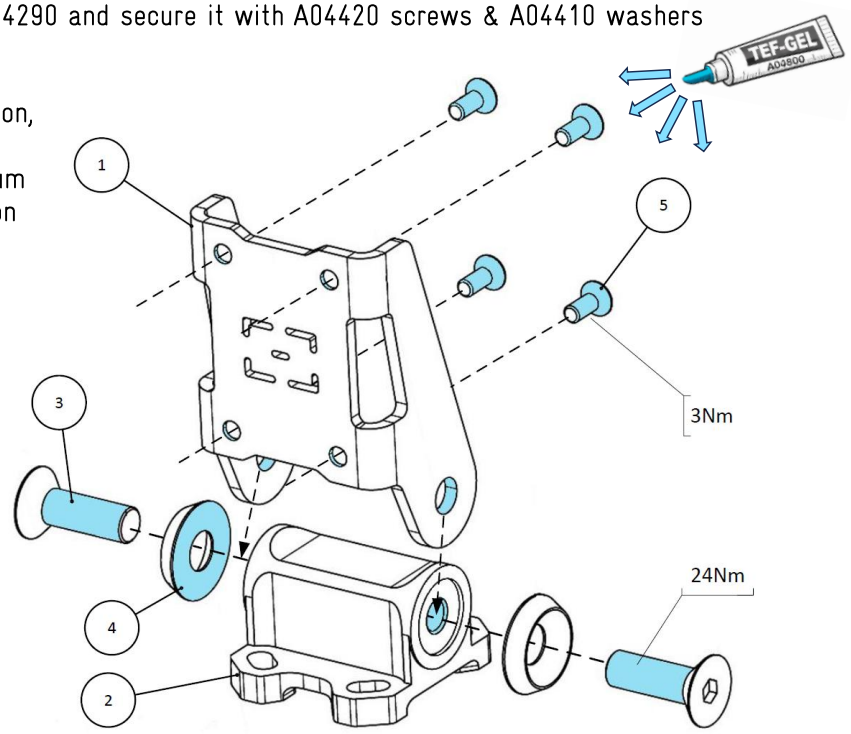
A4280 bracket's holes pattern in mm



3.10 Mounting the Watchkeeper on the Bracket

To attach the Watchkeeper system to the bracket A04280, use x4 screws A04430 (tightening torque 3Nm), then place the assembly on the bracket socket A04290 and secure it with A04420 screws & A04410 washers (tightening torque 24Nm).

To prevent the occurrence of galvanic corrosion, an insulating grease (Art N°A04800) must be applied to all contact areas between aluminum and stainless steel, areas identified in blue on the adjacent diagram.

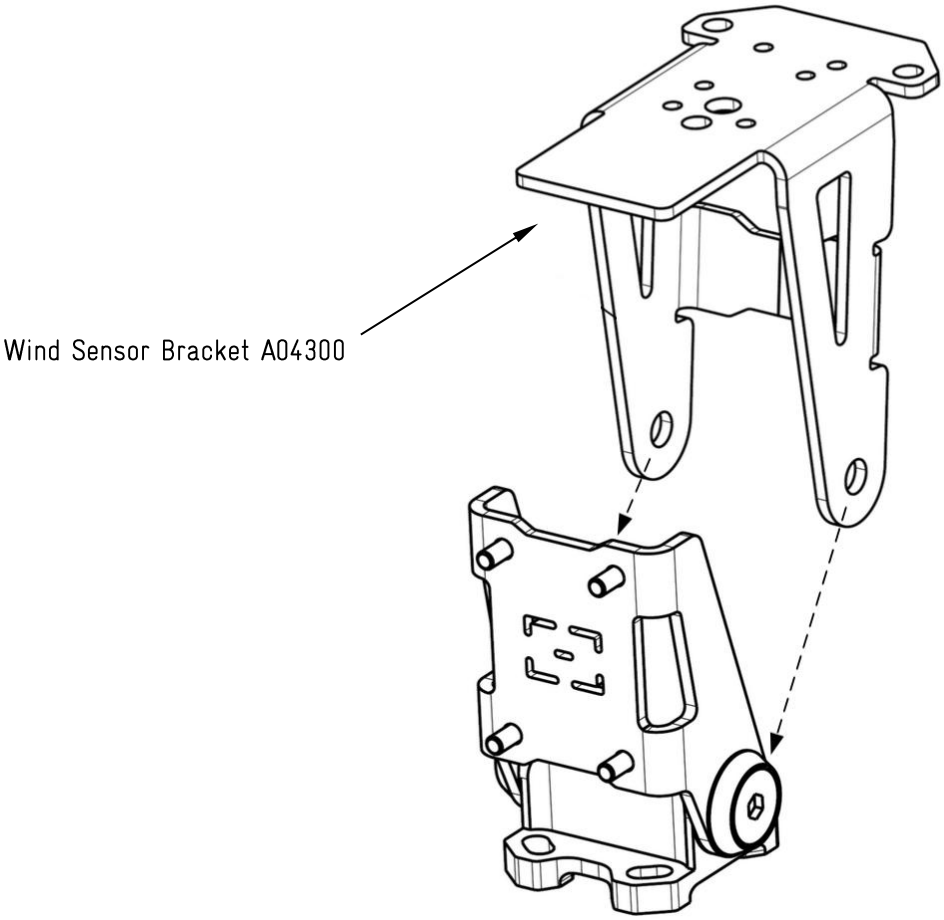


Description

1- A04280 Bracket
2- A04290 Bracket Socket
3- A04420 M8x25 Screw x2
4- A04410 Washer x2
5- A04430 M4x10 Screw x4

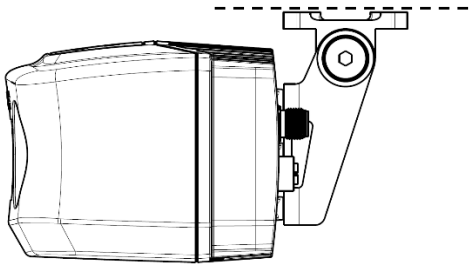
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NOTE : If required, the wind sensor bracket (A04300) can be attached to the bracket assembly using washers (A04410 & A04420) and M8 screws.

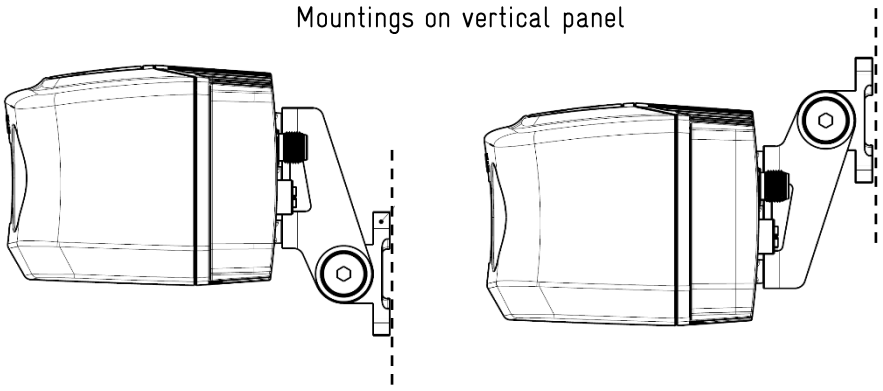


3.11 Alternative Mounting Options

Inverted mounting on horizontal panel

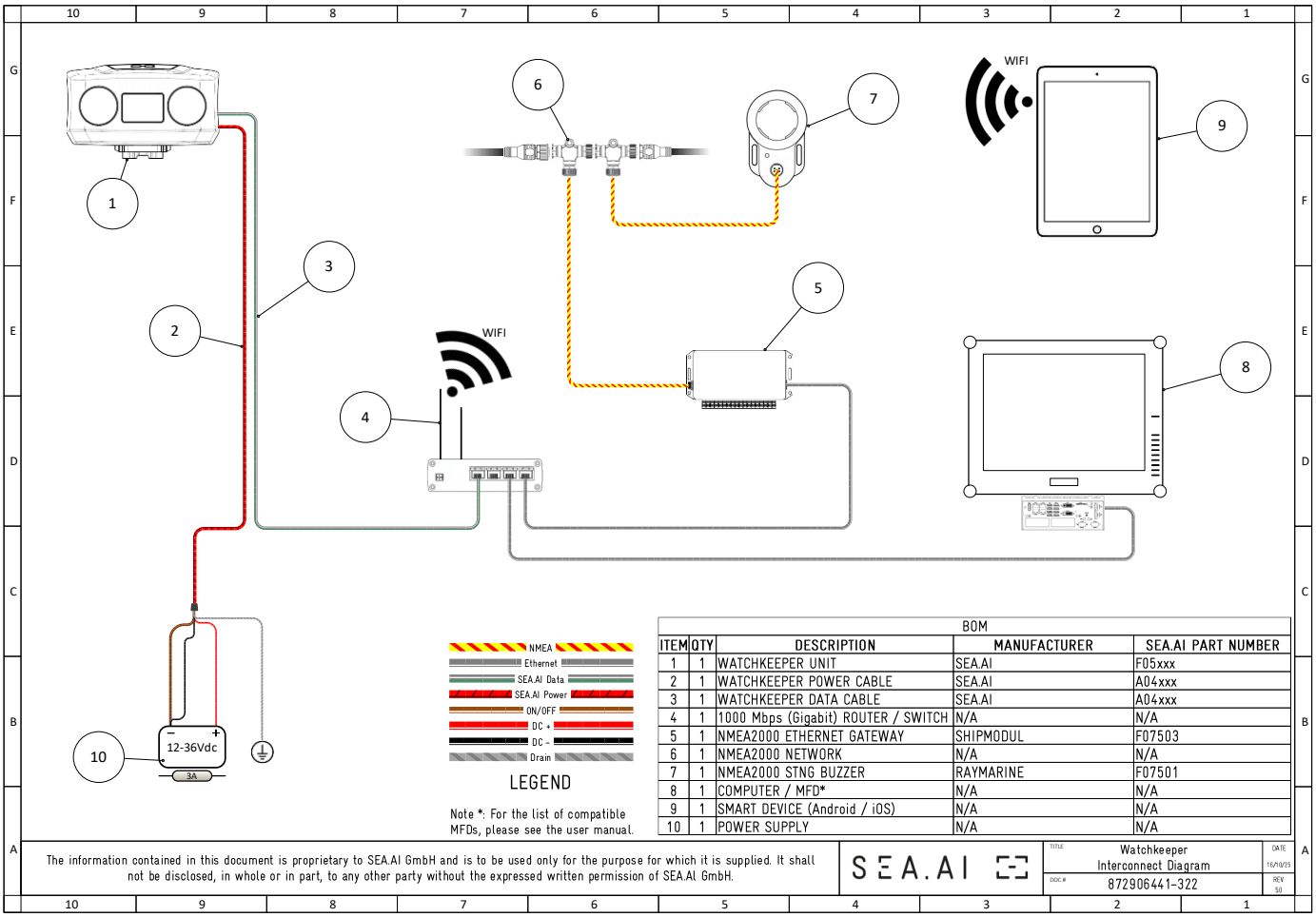


Mountings on vertical panel



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3.12 Interconnect Diagram



BOM

ITEM	QTY	DESCRIPTION	MANUFACTURER	SEA.AI PART NUMBER
1	1	WATCHKEEPER UNIT	SEA.AI	F05xxx
2	1	WATCHKEEPER POWER CABLE	SEA.AI	A04xxx
3	1	WATCHKEEPER DATA CABLE	SEA.AI	A04xxx
4	1	1000 Mbps (Gigabit) ROUTER / SWITCH	N/A	N/A
5	1	NMEA2000 ETHERNET GATEWAY	SHIPMODUL	F07503
6	1	NMEA2000 NETWORK	N/A	N/A
7	1	NMEA2000 STNG BUZZER	RAYMARINE	F07501
8	1	COMPUTER / MFD*	N/A	N/A
9	1	SMART DEVICE (Android / iOS)	N/A	N/A
10	1	POWER SUPPLY	N/A	N/A

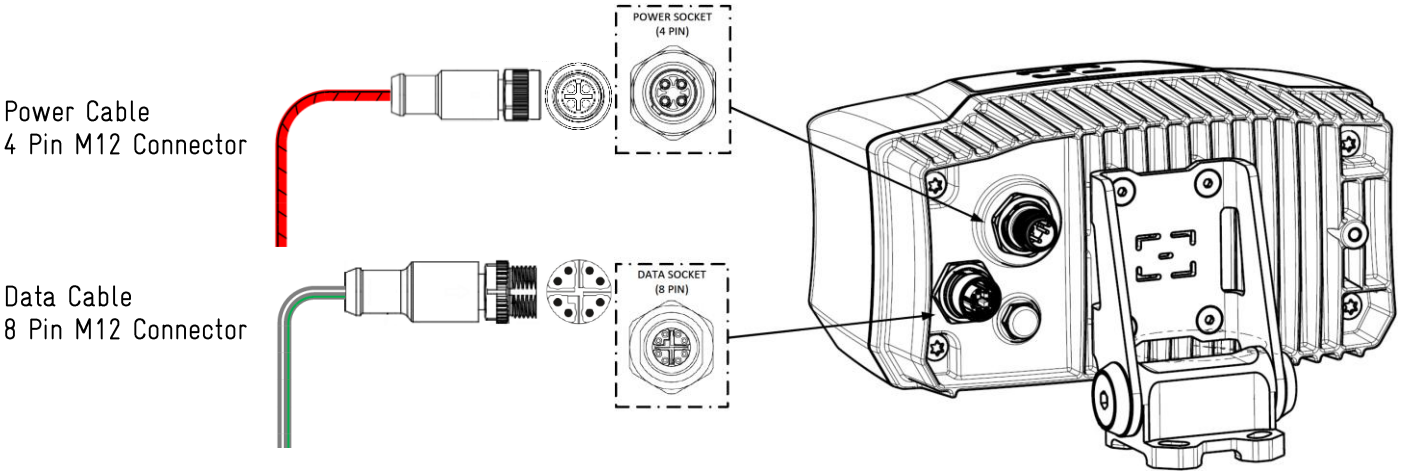
For the above drawing, please refer to: Watchkeeper Interconnect Diagram.

NOTE: SEA.AI does not supply items 4,6,8,9, or 10 but can recommend suitable manufacturers and models.

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3.13 Connecting Watchkeeper Unit

After routing the cables, connect and secure the data cable to the 8-pin socket and the power cable to the 4-pin socket at the rear of the system.



⚠ **WARNING:**

- Circular connectors must be fully secured.
- Cables must be properly supported so that their weight does not rest on the connector.

⚠ **CAUTION:**

- Use original SEA.AI cables and plugs, signal quality and performance will be reduced with incorrect cabling.

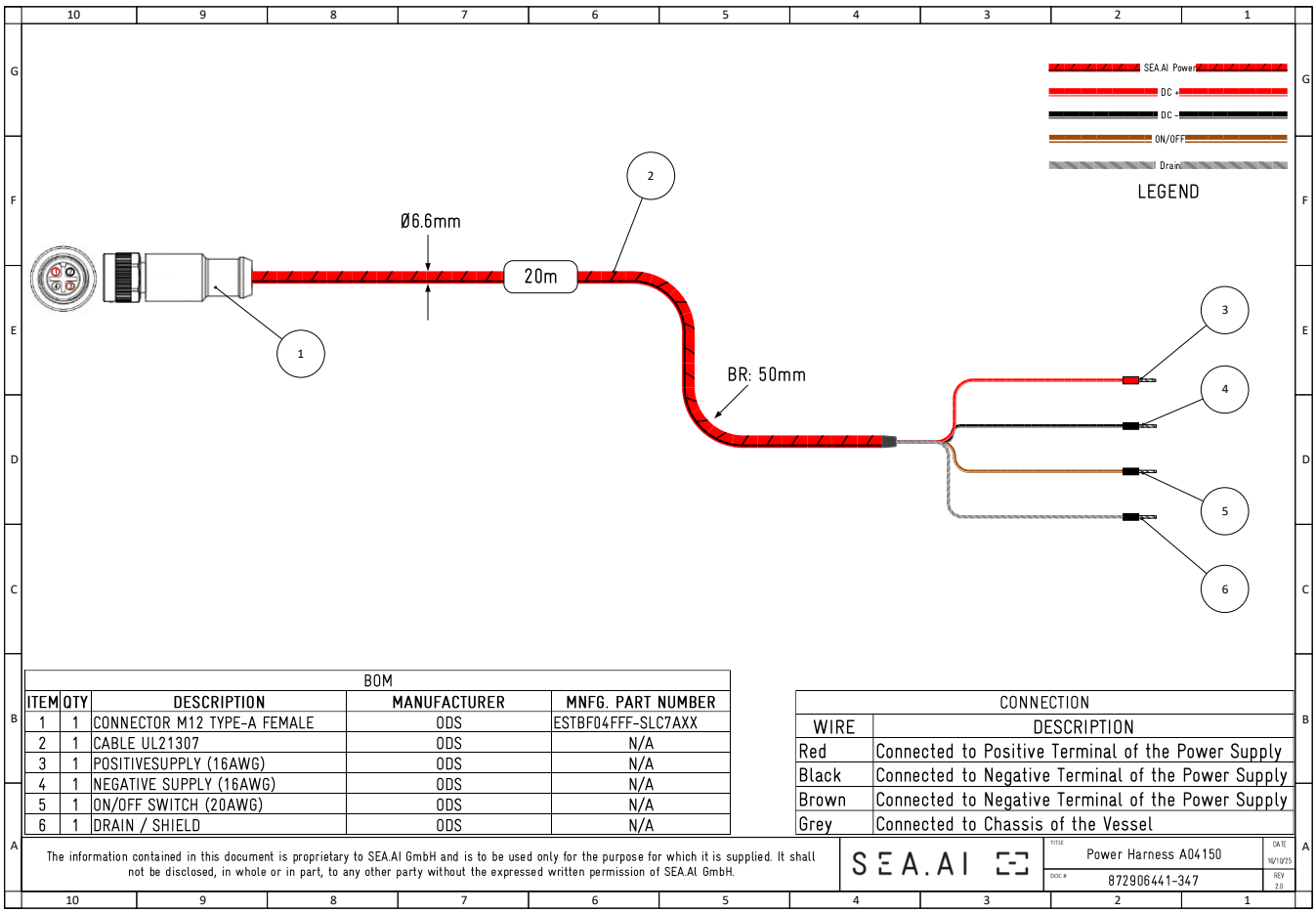
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3.14 Power Connection

Connect the power harness to the boat's electrical system:

- SEA.AI power, connects to 12V / 24V power supply.
- Red wire, connects to power supply's positive terminal.
- Black wire, connects to power supply's negative terminal.
- Brown wire, connects to power supply's negative terminal. Alternatively, it can be connected to a switch. System turns on when Brown wire is connected to ground.
- Drain wire (braid), connects to vessel's ground (chassis), or power supply's negative terminal.

3.15 Power Harness



BOM

ITEM	QTY	DESCRIPTION	MANUFACTURER	MNFG. PART NUMBER
1	1	CONNECTOR M12 TYPE-A FEMALE	ODS	ESTBF04FFF-SLC7AXX
2	1	CABLE UL21307	ODS	N/A
3	1	POSITIVESUPPLY (16AWG)	ODS	N/A
4	1	NEGATIVE SUPPLY (16AWG)	ODS	N/A
5	1	ON/OFF SWITCH (20AWG)	ODS	N/A
6	1	DRAIN / SHIELD	ODS	N/A

CONNECTION

WIRE	DESCRIPTION
Red	Connected to Positive Terminal of the Power Supply
Black	Connected to Negative Terminal of the Power Supply
Brown	Connected to Negative Terminal of the Power Supply
Grey	Connected to Chassis of the Vessel

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SEA.AI | TITLE: Power Harness A04150 | DOC #: 872906441-347 | DATE: 16/11/23 | REV: 2.0

For the above drawing, please refer to: Power Harness A04150.

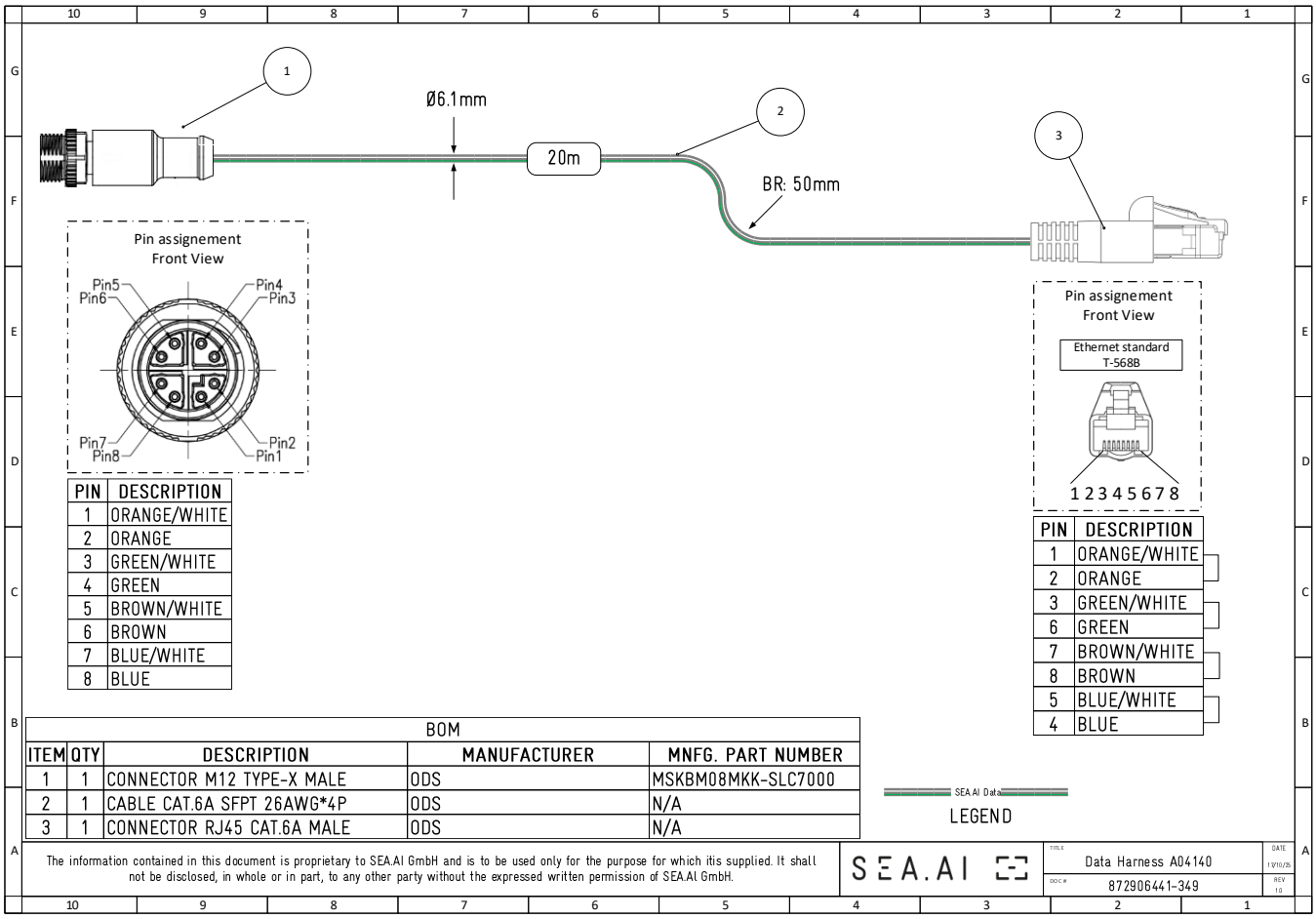
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3.16 Data Connection

Connect the data harness RJ45 connector to the vessel's LAN network – This may be a router, network switch, PC, or multifunctional display (MFD).

Note: Depending on the onboard network configuration and the make and model of the connected devices, a compatible RJ45 network adapter may be required.

3.17 Data Harness



For the above drawing, please refer to: Data Harness A04140.

NOTE: For optimal performance, the Watchkeeper requires a 1000 BASE-T (1000 Mbps bandwidth) connection.

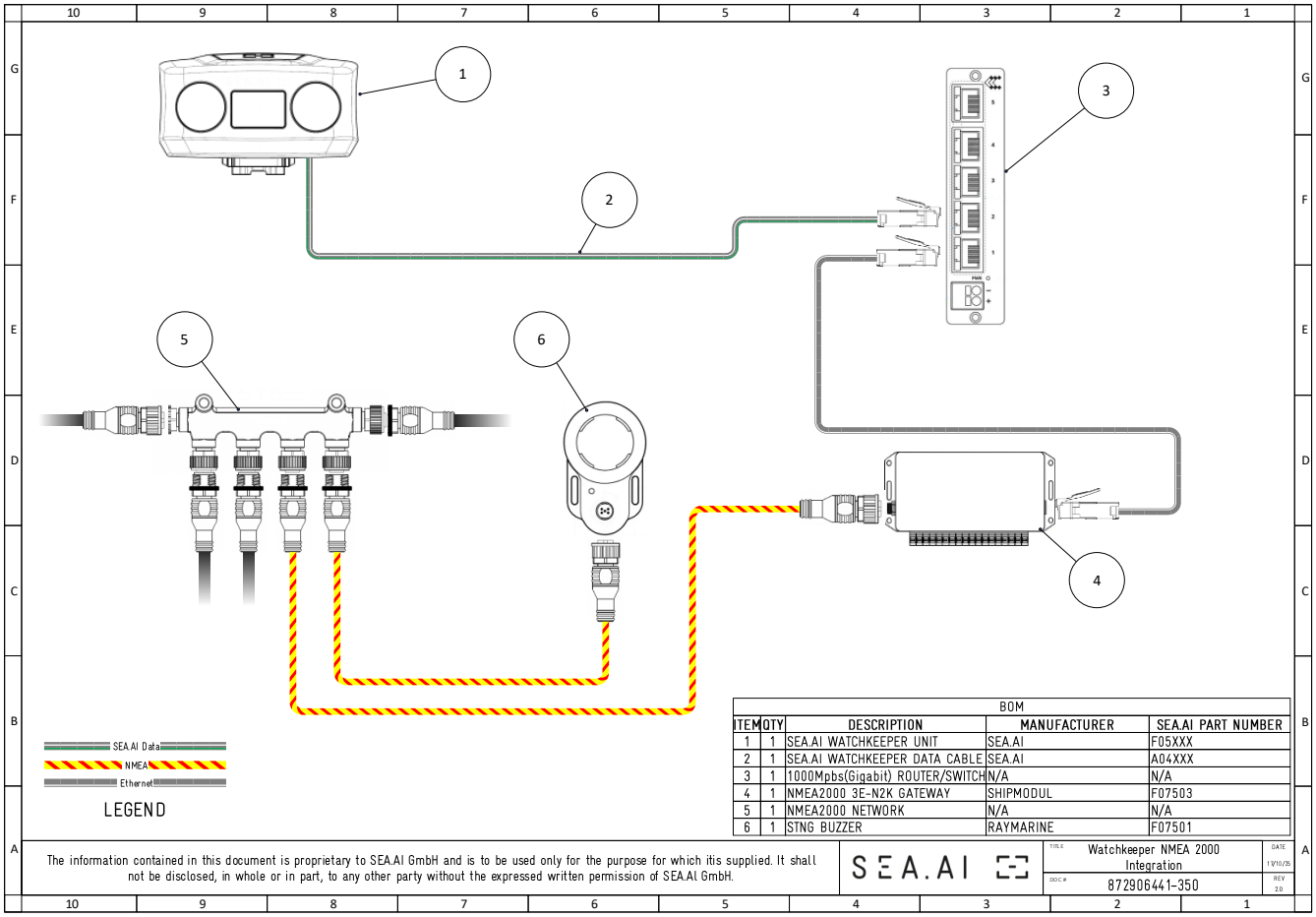
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3.18 Connecting to the NMEA 2000 Network

To enable external buzzer alarms and access additional vessel information from marine sensors (e.g., speed, GPS position, heading) via the SEA.AI Watchkeeper's Ethernet port, the vessel's NMEA 2000 bus data must be available on the LAN or Wi-Fi network using either the TCP/IP or UDP protocol.

Note: The core functionality of the SEA.AI Watchkeeper operates independently of NMEA 2000 integration. However, connecting the Watchkeeper to the vessel network via an Ethernet Gateway is recommended to unlock future features.

The NMEA2000 Ethernet Gateway Shipmodul MiniPlex-3E-N2K (F07503) is supported. For detailed instructions on configuring the N2K Gateway, please refer to the Shipmodul MiniPlex 3E-N2K user manual: https://www.shipmodul.com/miniplex-3e-n2k.html



BOM

ITEM	QTY	DESCRIPTION	MANUFACTURER	SEA.AI PART NUMBER
1	1	SEA.AI WATCHKEEPER UNIT	SEA.AI	F05XXX
2	1	SEA.AI WATCHKEEPER DATA CABLE	SEA.AI	A04XXX
3	1	1000Mpbs(Gigabit) ROUTER/SWITCH	N/A	N/A
4	1	NMEA2000 3E-N2K GATEWAY	SHIPMODUL	F07503
5	1	NMEA2000 NETWORK	N/A	N/A
6	1	STNG BUZZER	RAYMARINE	F07501

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SEA.AI

TITLE: Watchkeeper NMEA 2000 Integration
DOC #: 872906441-350
DATE: 1/9/10/25
REV: 2.0

NOTE: Items 4 and 6 (not included in the installation kit) are available as optional accessories. SEA.AI does not supply items 3 and 5 but can recommend suitable manufacturers and models.

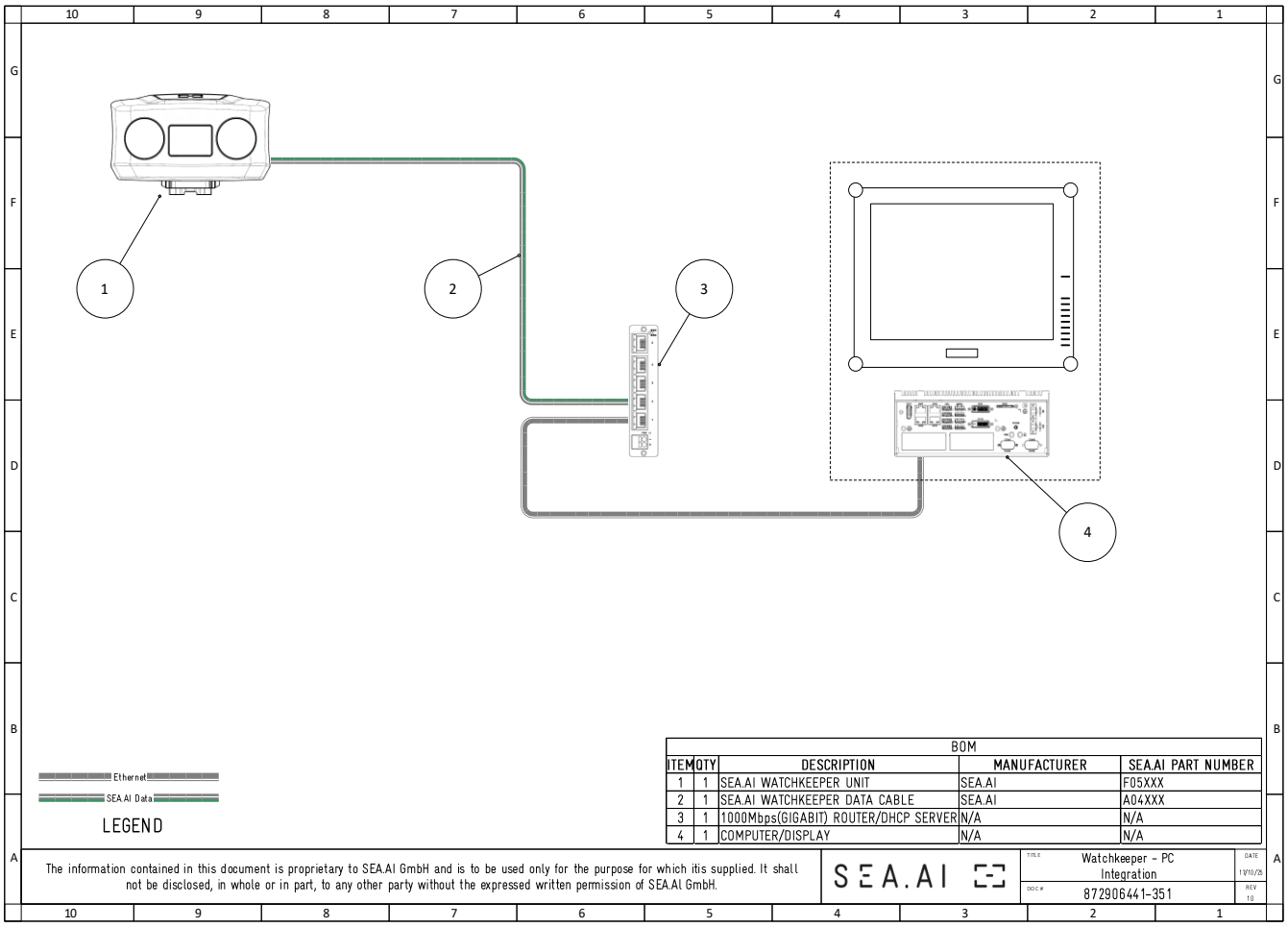
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4 Integration Scenarios

The integration of the SEA.AI Watchkeeper system may vary depending on the configuration of the onboard equipment and the type of available interfaces. The most common integration scenarios are described below.

4.1 PC Integration

For PC integrations, a DHCP server or router must be available on the network to enable communication between the Watchkeeper system and the PC.



BOM

ITEM	QTY	DESCRIPTION	MANUFACTURER	SEA.AI PART NUMBER
1	1	SEA.AI WATCHKEEPER UNIT	SEA.AI	F05XXX
2	1	SEA.AI WATCHKEEPER DATA CABLE	SEA.AI	A04XXX
3	1	1000Mbps(GIGABIT) ROUTER/DHCP SERVER	N/A	N/A
4	1	COMPUTER/DISPLAY	N/A	N/A

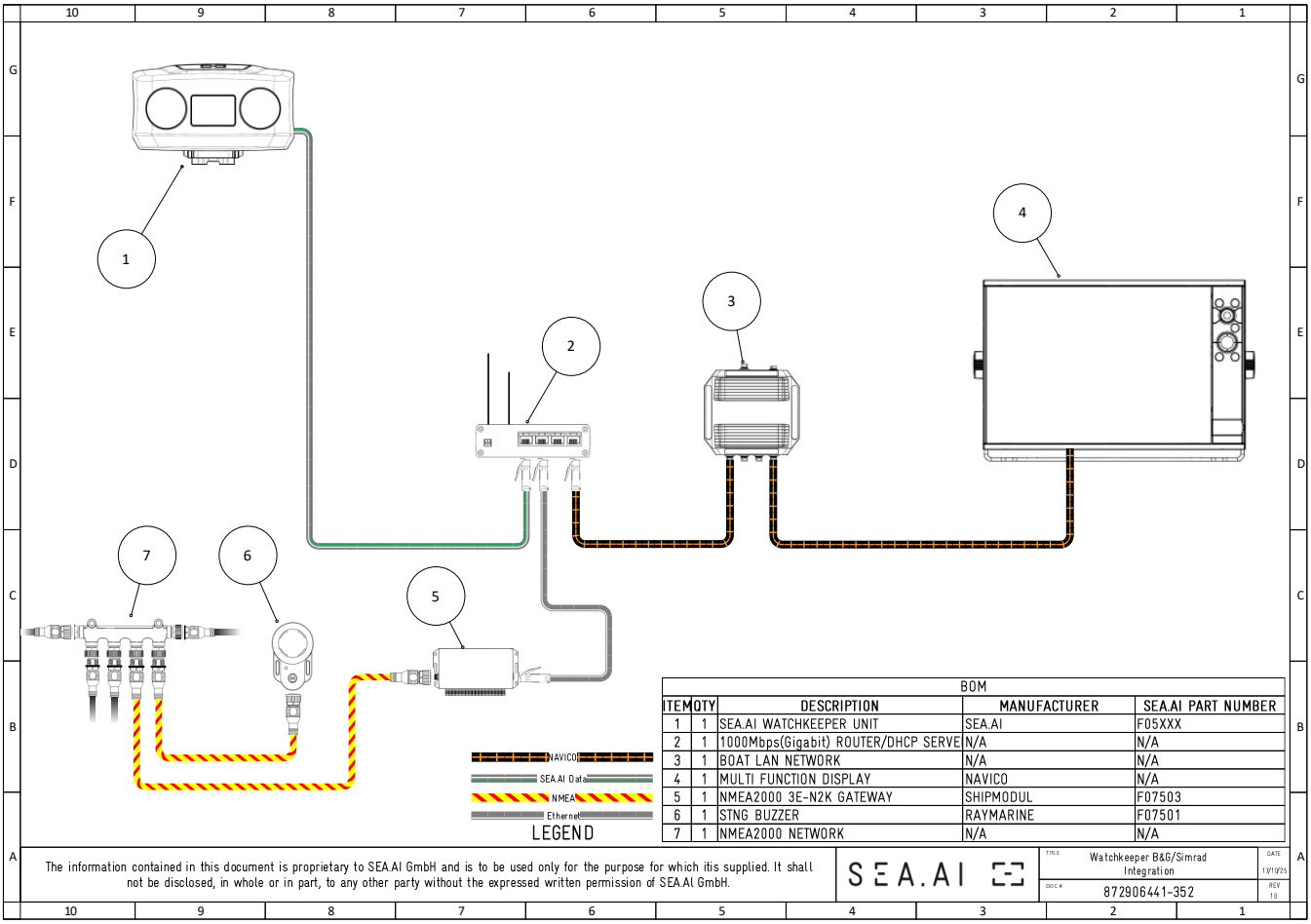
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SEA.AI | TITLE: Watchkeeper - PC Integration | DOC #: 872906441-351 | DATE: 1/9/25 | REV: 1.0

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4.2 B&G®/Simrad® Integration

When integrating the Watchkeeper system with B&G®/Simrad® multi-functional displays (MFDs), a router providing DHCP server functionality is required to facilitate communication with the connected MFDs.



BOM

ITEM	QTY	DESCRIPTION	MANUFACTURER	SEA.AI PART NUMBER
1	1	SEA.AI WATCHKEEPER UNIT	SEA.AI	F05XXX
2	1	1000Mbps(Gigabit) ROUTER/DHCP SERVE	N/A	N/A
3	1	BOAT LAN NETWORK	N/A	N/A
4	1	MULTI FUNCTION DISPLAY	NAVICO	N/A
5	1	NMEA2000 3E-N2K GATEWAY	SHIPMODUL	F07503
6	1	STNG BUZZER	RAYMARINE	F07501
7	1	NMEA2000 NETWORK	N/A	N/A

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SEA.AI

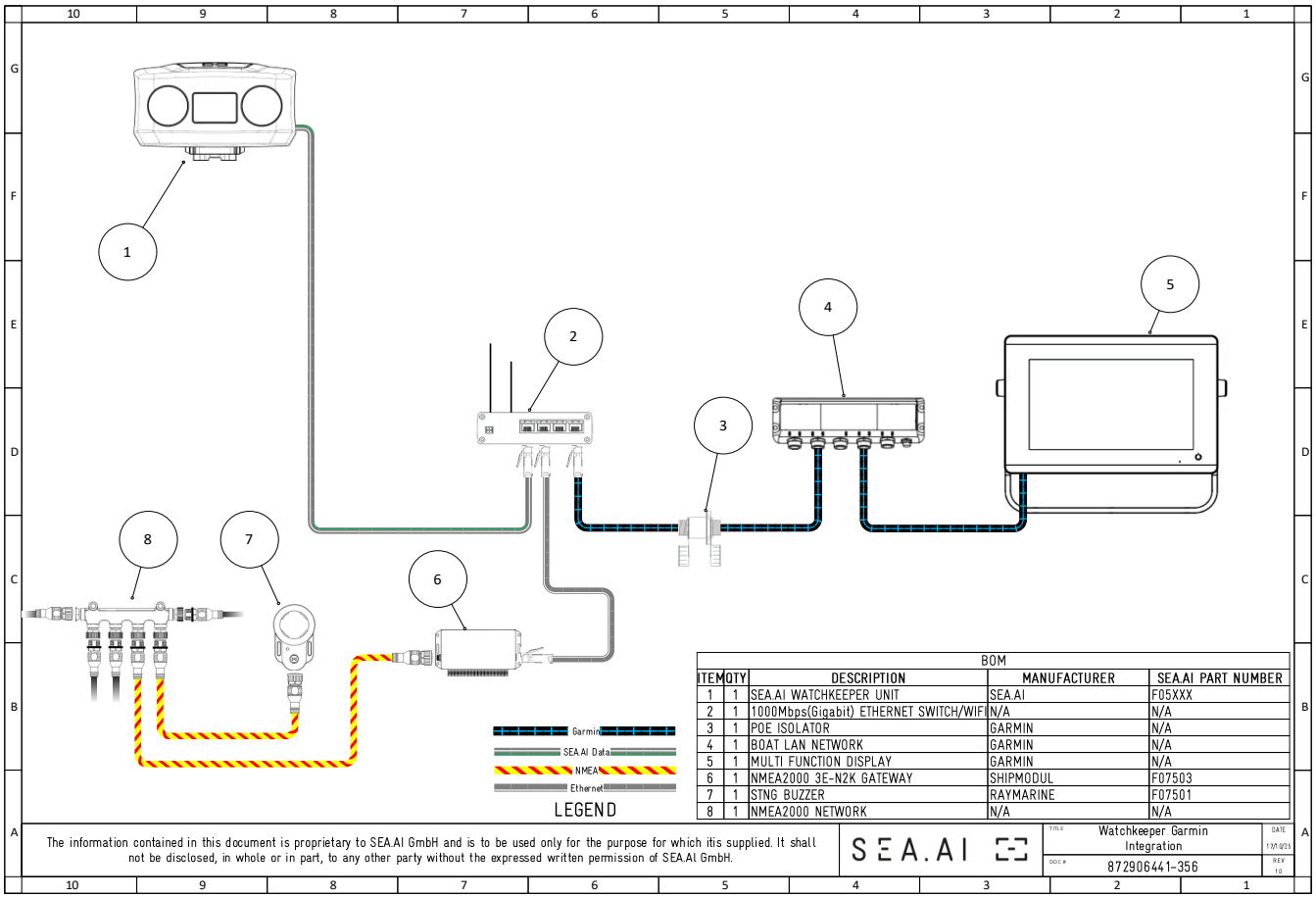
TITLE: Watchkeeper B&G/Simrad Integration
DOC.#: 872906441-352
DATE: 19/10/25
REV: 1.0

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4.3 Garmin® Integration

When integrating with Garmin® multi-functional displays (MFDs), install a **PoE isolator**, provided by Garmin, on the Garmin network switch port connected to the Watchkeeper.

If mobile devices are to be connected, a Wi-Fi router may be added to the network; ensure that its **DHCP function is disabled** to avoid IP address conflicts with the MFD.



BOM

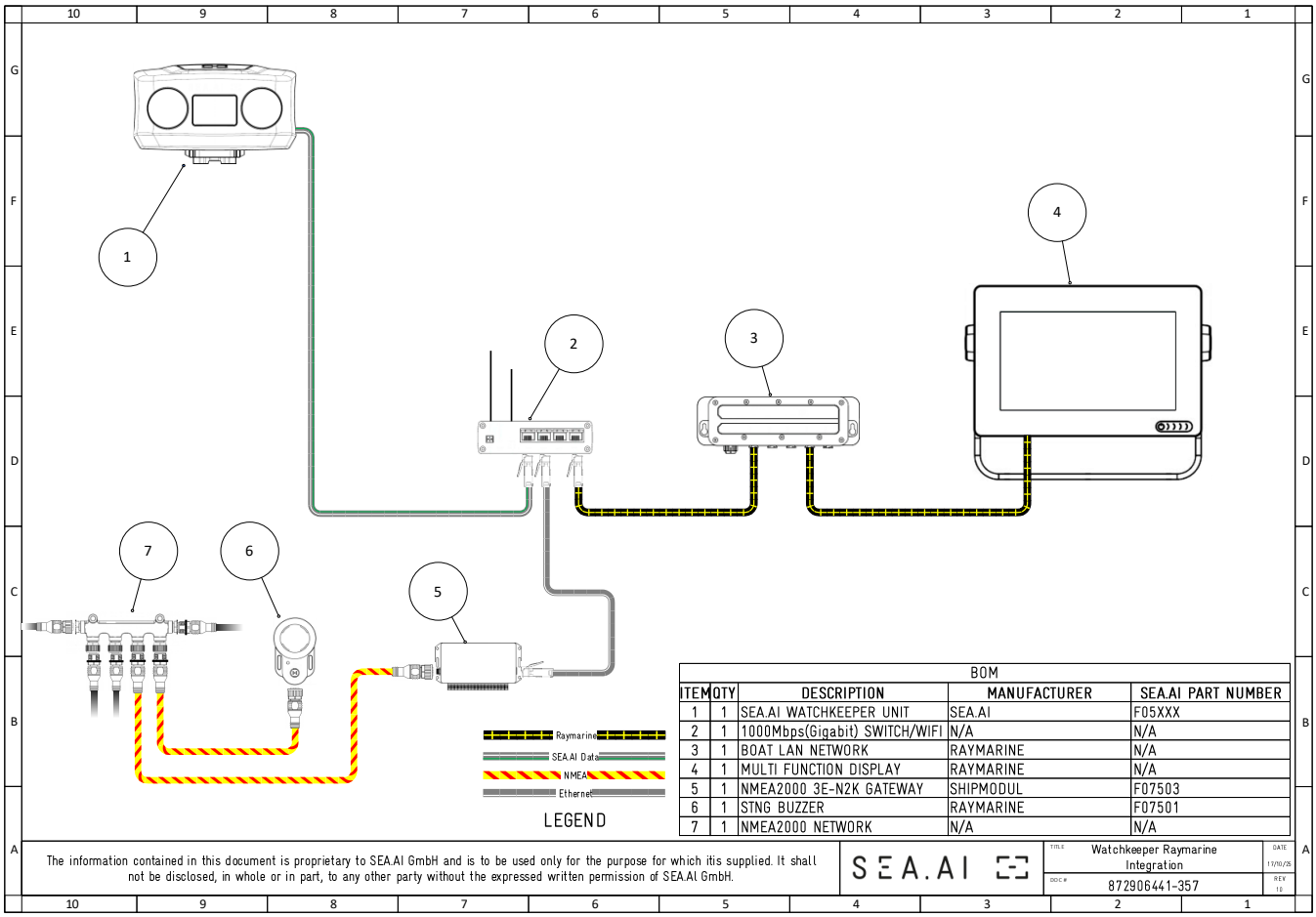
ITEM	QTY	DESCRIPTION	MANUFACTURER	SEA.AI PART NUMBER
1	1	SEA.AI WATCHKEEPER UNIT	SEA.AI	F05XXX
2	1	1000Mbps(Gigabit) ETHERNET SWITCH/WIFI	N/A	N/A
3	1	POE ISOLATOR	GARMIN	N/A
4	1	BOAT LAN NETWORK	GARMIN	N/A
5	1	MULTI FUNCTION DISPLAY	GARMIN	N/A
6	1	NMEA2000 3E-N2K GATEWAY	SHIPMODUL	F07503
7	1	STNG BUZZER	RAYMARINE	F07501
8	1	NMEA2000 NETWORK	N/A	N/A

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SEA.AI | TITLE: Watchkeeper Garmin Integration | DOC #: 872906441-356 | DATE: 1/7/2025 | REV: 1.0

4.4 Raymarine® Integration

When integrating with a Raymarine® multi-functional displays (MFDs), a Wi-Fi router may be installed to enable mobile device access. Ensure that the router's **DHCP function is turned off** to avoid IP address conflicts with the MFD.



BOM

ITEM	QTY	DESCRIPTION	MANUFACTURER	SEA.AI PART NUMBER
1	1	SEA.AI WATCHKEEPER UNIT	SEA.AI	F05XXX
2	1	1000Mbps(Gigabit) SWITCH/WIFI	N/A	N/A
3	1	BOAT LAN NETWORK	RAYMARINE	N/A
4	1	MULTI FUNCTION DISPLAY	RAYMARINE	N/A
5	1	NMEA2000 3E-N2K GATEWAY	SHIPMODUL	F07503
6	1	STNG BUZZER	RAYMARINE	F07501
7	1	NMEA2000 NETWORK	N/A	N/A

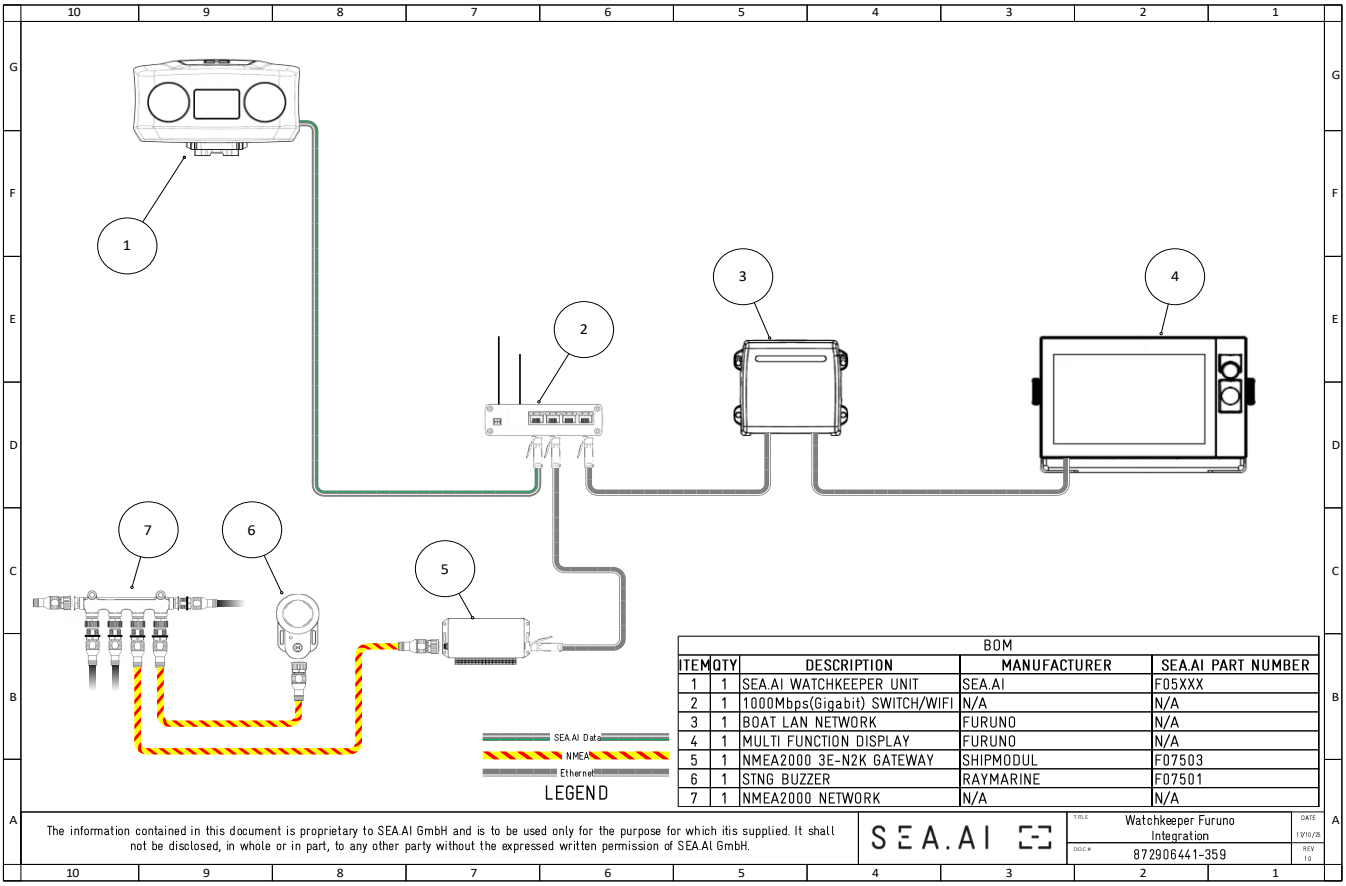
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SEA.AI | TITLE: Watchkeeper Raymarine Integration | DATE: 17/10/25 | DOC #: 872906441-357 | REV: 1.0

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4.5 Furuno® Integration

When integrating with a Furuno® multi-functional displays (MFDs), a Wi-Fi router may be installed to enable mobile device access. Ensure that the router's **DHCP function is turned off** to avoid IP address conflicts with the MFD.



BOM

ITEM	QTY	DESCRIPTION	MANUFACTURER	SEA.AI PART NUMBER
1	1	SEA.AI WATCHKEEPER UNIT	SEA.AI	F05XXX
2	1	1000Mbps(Gigabit) SWITCH/WIFI	N/A	N/A
3	1	BOAT LAN NETWORK	FURUNO	N/A
4	1	MULTI FUNCTION DISPLAY	FURUNO	N/A
5	1	NMEA2000 3E-N2K GATEWAY	SHIPMODUL	F07503
6	1	STNG BUZZER	RAYMARINE	F07501
7	1	NMEA2000 NETWORK	N/A	N/A

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SEA.AI

TITLE: Watchkeeper Furuno Integration
DOC #: 872906441-359
DATE: 19/10/25
REV: 1.0

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5 SYSTEM SETUP

5.1 First start up

Power on the SEA.AI Watchkeeper by switching on the dedicated circuit breaker that supplies power to the system (refer to the *Power Connection* chapter for wiring details).

> NOTE: System start-up, including loading the SEA.AI software, may take up to 3 minutes. The user interface will become available once the start-up process is complete.

Switch on all required accessories, such as the PC, router, network switch, tablet and any connected interface (e.g. display, MFD, keyboard, etc.).

5.1.1 Accessing the SEA.AI User Interface via PC

You can access the SEA.AI Watchkeeper user interface using one of the following methods:

Option 1 – SEA.AI Watchkeeper Application (Recommended):

Download the SEA.AI Watchkeeper Windows, macOS, or Linux application from the SEA.AI Document Library (Watchkeeper Document Library).

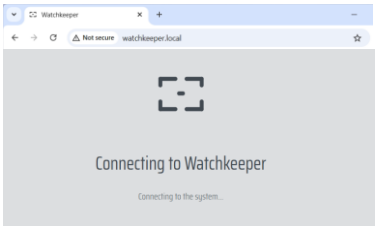
Open the SEA.AI application.

If your PC is connected to the same network as the Watchkeeper, it will automatically detect the system

Option 2 – Web Browser Access:

Open a web browser.

Enter the following address in the address bar: http://watchkeeper.local/



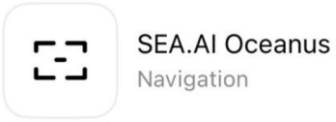
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5.1.2 Accessing the SEA.AI User Interface via Mobile Device

1. Download the latest version of the SEA.AI application:

App Store: SEA.AI Oceanus





Google Play Store: SEA.AI





2. Connect your mobile device to the Wi-Fi network that transmits SEA.AI data (e.g. router or MFD)
3. Launch the SEA.AI application to access the user interface.

5.1.3 Accessing the SEA.AI User Interface via a Multi-Functions Display (MFD)

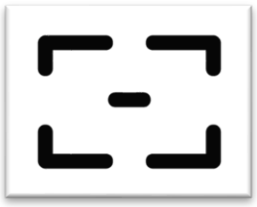
Performance Notice:

The computing capability of your MFD may limit the performance or visual quality of the SEA.AI application. To ensure optimal operation, the system automatically adapts to your MFD's brand and model. Connection methods and interface appearance may vary accordingly.
For detailed compatibility information, refer to the **Watchkeeper MFD Compatibility List** on **page 28** of this

- Launching the SEA.AI Application on your MFD:

  1. On your MFD, navigate to Third-Party Apps.

  2. Tap the SEA.AI icon. The icon appears automatically once the Watchkeeper is powered on and connected to the network.

SEA.AI Icon 

- Accessing RTPS Video Streams:

  To view the RTSP video stream directly on your MFD, refer to your MFD's user manual for instructions on setting up RTSP streams.

  The **Watchkeeper RTSP address** is available on **page 34** of this manual.

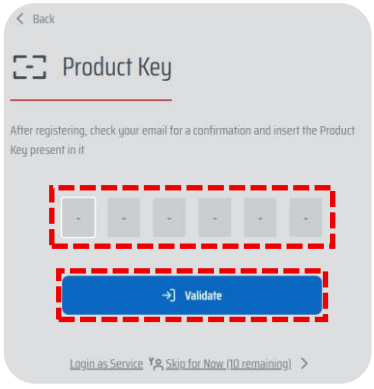
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5.2 First-Time Activation

When starting the system for the first time, you will be prompted to enter the digital activation key provided during product registration (see *"Product Registration, Activation and Warranty"* on page 7).

1. Click **Insert Product Key**.
2. Enter the **6-character activation key**.
3. Click **Validate** to complete the activation process.





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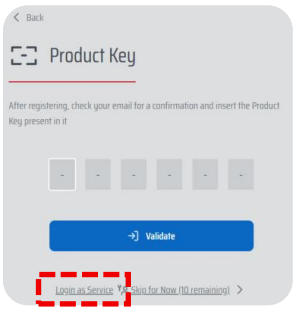
5.3 Installer mode

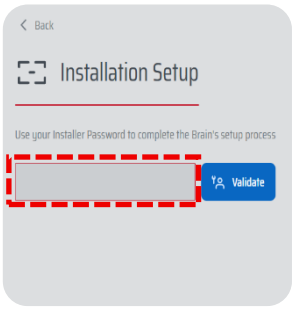
If the system has not been registered yet, **Installer mode** provides access to all functionalities for installation, configuration, and maintenance.

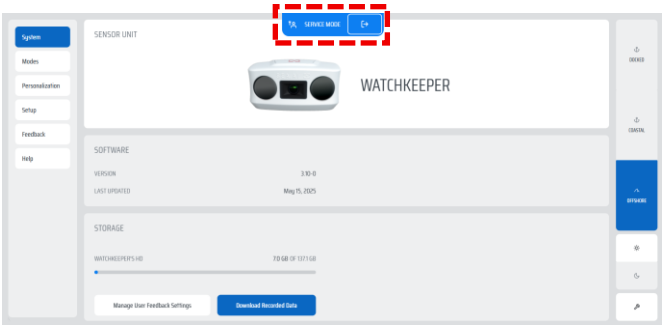
Installer mode functionality is accessible via Watchkeeper registration page.

1. Click **Insert Product Key**.
2. Enter the **Login as Service**.
3. Enter the **Installer Password**. Please contact Service for the password.
4. Click **Validate** to Login.
5. Upon successful Login, there will be **SERVICE MODE** displayed on top of the user interface.
6. To exit Installer mode, click the service mode arrow.









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5.4 Setting Parameters

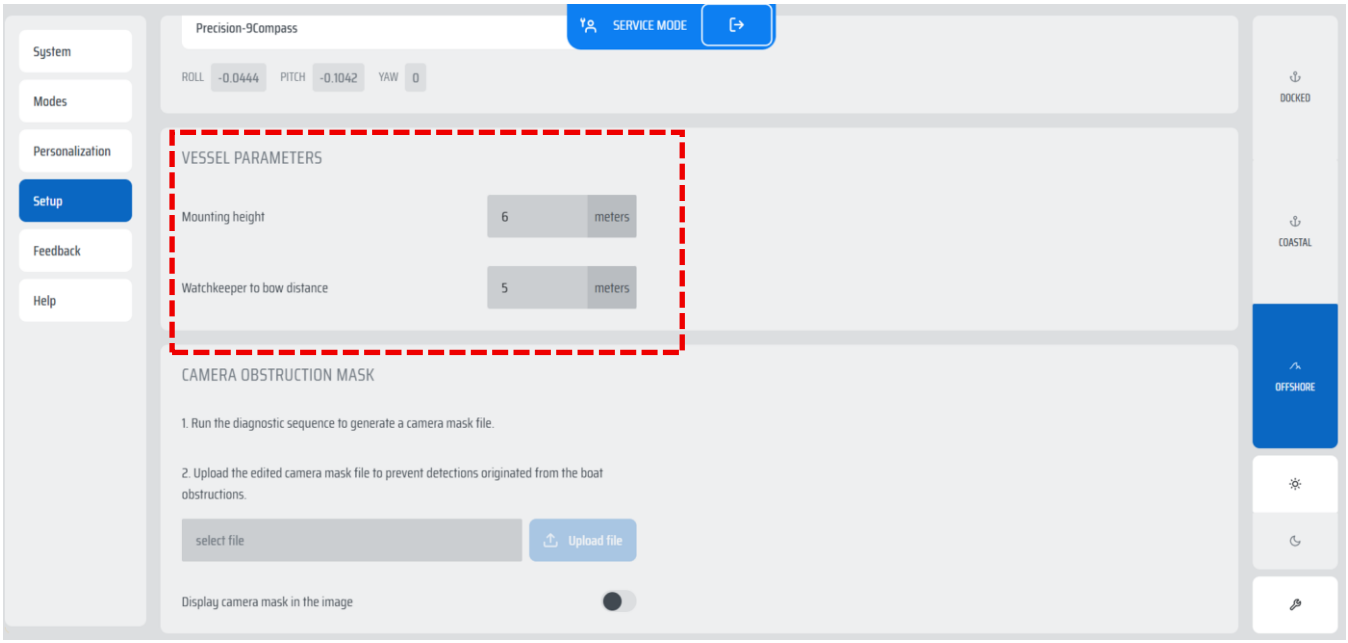
For optimal alarm performance it is essential to correctly configure the mounting height and distance from the unit to the bow.

Note:

- *Mounting height* is the vertical distance (in meters) between the water surface and the Watchkeeper unit.

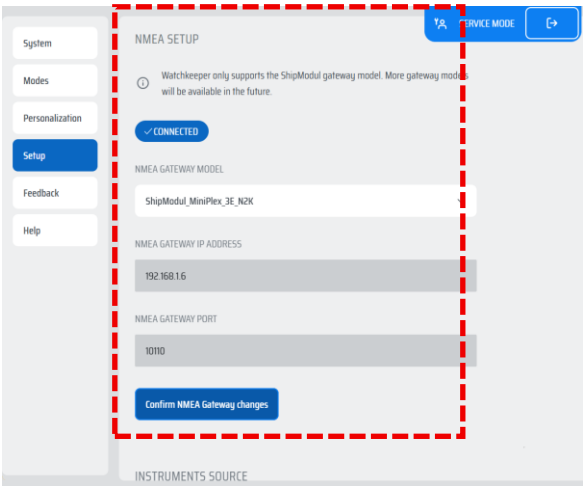
To set these parameters:

1. Open the **Settings** menu.
2. Select the **Setup** tab.
3. Scroll down to the **Installer Login** section.



5.5 Setting Up the NMEA 2000 Gateway

1. Select "**Shipmodul_MiniPlex_3E_N2K**" as NMEA Gateway Model.
2. Enter the **NMEA Gateway IP Address** (e.g.192.168.1.6).
3. Enter the **NMEA Gateway Port** (e.g.10110).
4. Click **Apply NMEA Gateway Changes**.
5. Once the connection is established, the status indicator turns blue and displays "CONNECTED"



For detailed instructions on configuring NMEA 2000 Gateway, please refer to Shipmodul Miniplex 3E-N2K User Manual available at: https://www.shipmodul.com/miniplex-3e-n2k.html.

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5.6 Motor Yacht - Vertical Angle Adjustment

The vertical angle of the Watchkeeper system must be adjusted for each vessel individually. This setting depends on the vessel's trim – its position in the water when sailing at cruising speed.

To minimize the blind spot in front of the bow at cruising speed:

1. Open the Thermal View on the display.
2. Adjust the vertical angle so that the horizon line appears approximately at three-quarters (¾) of the height of the onboard display

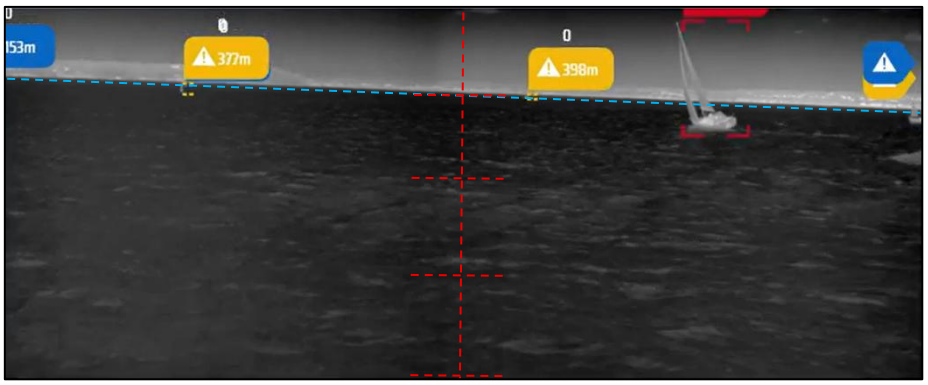
System adjusted while the vessel is stationary in harbour – results in a large blind spot in front of the bow:



System adjusted according to the vessel's angle at cruising speed – **minimizes the blind spot** in front of the bow.



To verify correct adjustment:
During navigation, the **horizon line** in the **Thermal View** should appear at approximately **three-quarters (¾)** of the height of the onboard display. See the guidance below:



NOTE: The blue and red dotted lines shown in the illustration are for instructional purposes only and do not appear in the actual SEA.AI application.

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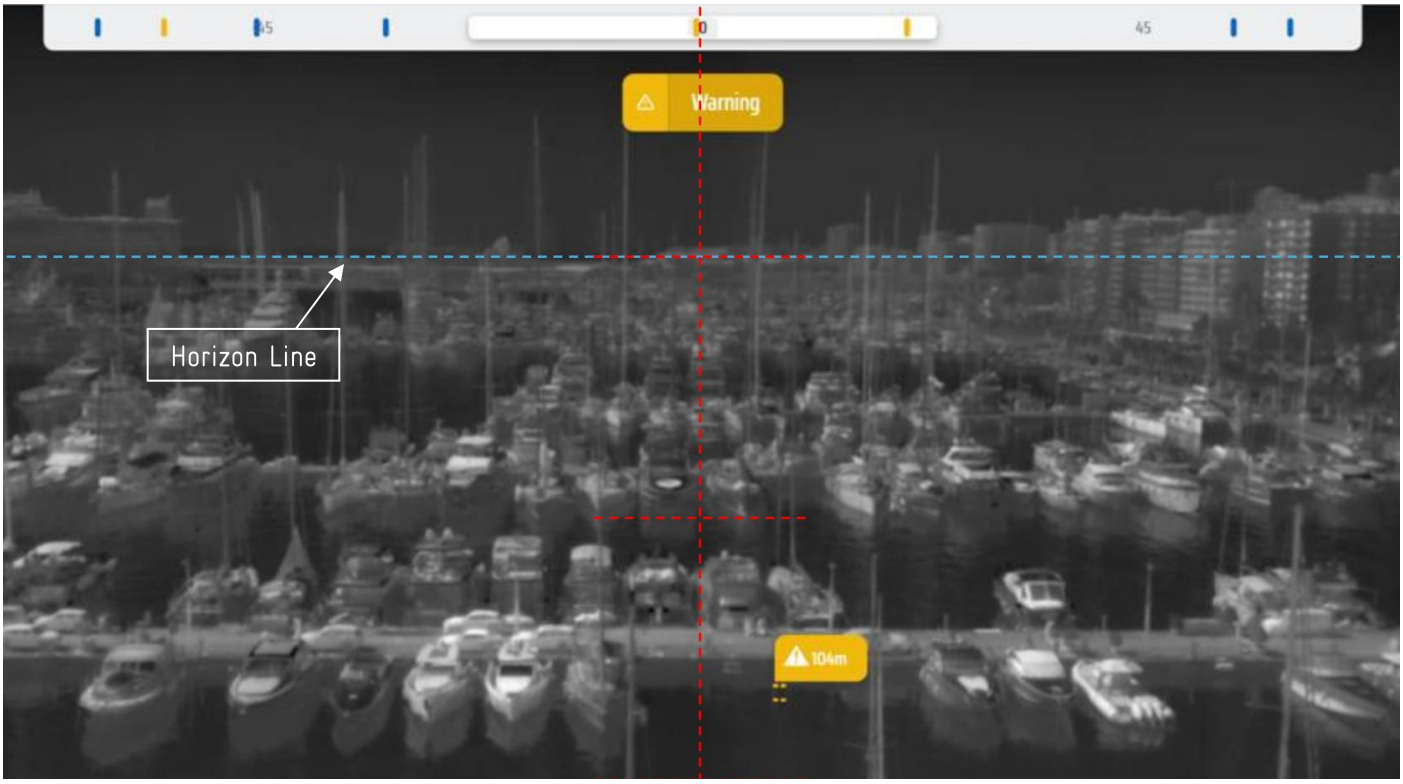
5.7 Sailing Yacht – Vertical Angle Adjustment

Due to their higher installation position, sailing yacht systems are subject to greater vertical angle variations when crossing waves. To maintain a clear view of the horizon and ensure optimal water surface analysis performance, a proper vertical alignment adjustment is required.

Using the Thermal View, adjust the vertical angle so that the horizon line appears at approximately two-thirds (⅔) of the height of the onboard display.

This configuration helps stabilize the operator's visual reference and enhances analysis accuracy under varying sea conditions.

To verify correct adjustment:
During navigation, the horizon line in the Thermal View should appear at approximately two-thirds (⅔) of the height of the onboard display. See the guidance below:



NOTE: The blue and red dotted lines shown in the illustration are for instructional purposes only and do not appear in the actual SEA.AI application.

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5.8 Ignore Mask

Due to the environments in which Watchkeeper systems are installed, there may be visual obstructions within the system's field of view that can cause **false positive alarms**. These obstructions may include structural elements on the vessel, such as antennas, masts, or rigging.

To prevent false detections, these obstructions must be captured and stored in the system's memory so they can be excluded from the AI-based detection algorithms. This process is performed by creating a Camera Obstruction Mask.

For detailed instructions on creating an obstruction mask, please refer to the Watchkeeper Ignore Mask Creation Guideline available in the Watchkeeper Document Library.

Example: Boat structure visible in the RGB field of view requiring the creation of an obstruction mask to prevent false detections caused by this structure.



Example: The same vessel with the RGB ignore mask applied (areas displayed in red).



⚠ **NOTICE** – An improperly configured mask may cause false detections or reduced system performance. Please recreate the mask or contact service@sea.ai for assistance.

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6 WATCHKEEPER SOFTWARE UPDATE

At SEA.AI, we are committed to providing the best possible user experience by regularly releasing software updates. These updates deliver the latest technological improvements, feature enhancements, and critical bug fixes to ensure optimal system performance.

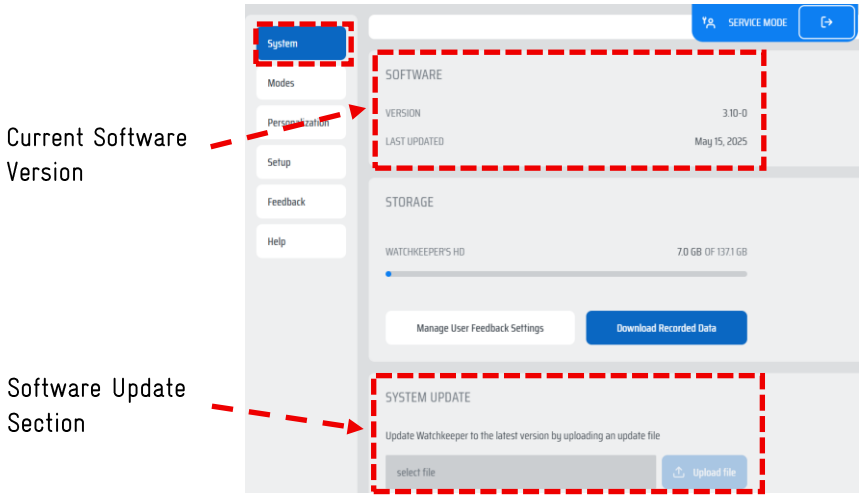
Minimum Requirements

Before performing a software update, ensure the following:

- SEA.AI Watchkeeper System
- Computer
- Internet Connection
- Watchkeeper and the Computer are on the same Local Area Network (LAN)

Step-by-Step Software Update Procedure

1. Access the software download page: Visit the www.sea.ai website, navigate to the Document library, and select Watchkeeper. Further navigate to Watchkeeper Software folder.
2. Download the update file: Download the file names SEA-AI_WK_x.x-xx.oscar (~180MB in size) to a known folder on your PC.
3. Open the Watchkeeper User Interface:
    - Launch the SEA.AI Desktop App, or
    - Open a web browser (e.g. Chrome) and enter http://watchkeeper.local/
4. Navigate to system settings: In the main User Interface, open the Settings menu
5. Check the current software version: Select the System tab and scroll down to the Software section. Note the current software version for later verification.
6. Upload the update package: Scroll to the System Update section and click Select File.
7. Choose the update package: When prompted, navigate to the location of the update file downloaded in Step 2 and select it.
8. Start the update: Click Upload File and monitor the update process on the screen.
9. Verify the update: After the system restarts, return to Settings -> System -> Software and confirm that the new software version matches the latest release.
10. Update completed: The software update is now successfully completed.



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7 MFD COMPATIBILITY

The table below lists the features available for major MFD models (note: this list is not exhaustive).

For more information or to confirm compatibility with other models, please contact your regional SEA.AI Sales Manager.

SEA.AI

Watchkeeper MFD Compatibility (As of September 2025)

Brand	Model	Sea.AI App	RTSP Streams & Sea.AI App	Monitor Mode for mini PC (HDMI Connection)	Alarms
Raymarine	Axiom+ Series	✓	✗	✗	MFD Alarm
	Axiom Pro Series	✓	✗	✗	MFD Alarm
	Axiom 2 Pro Series	✓	✗	✗	MFD Alarm
	Axiom XL Series	✓	✗	✓	MFD Alarm
	Axiom 2 XL Series	✓	✗	✓	MFD Alarm
GARMIN	GPSMAP 9xxx Series	✓	✗	✓	NMEA Buzzer
	GPSMAP 84xx Series	✓	✗	✓	NMEA Buzzer
navico B&G LOWRANCE SIMRAD	Zeus 3S	✗	✓	✗	MFD Alarm
	NSS EVO3S	✗	✓	✗	MFD Alarm
	NSO EVO3S	✗	✓	✓	MFD Alarm
	NSX	✓	✗	✗	MFD Alarm
	Zeus S	✓	✗	✗	MFD Alarm
	Zeus SR	✓	✗	✗	MFD Alarm
FURUNO	TZT 16F	✗	✓ Lower frame rate and/or resolution	✓	NMEA Buzzer
	TZT 19F	✗	✓ Lower frame rate and/or resolution	✓	NMEA Buzzer
	TZT 2BB	✗	✓ Lower frame rate and/or resolution	✓	NMEA Buzzer
	TZT 16-22-24X	✗	✓ Lower frame rate and/or resolution	✓	NMEA Buzzer
	TZT BBX	✗	✓ Lower frame rate and/or resolution	✓	NMEA Buzzer

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8 RTSP VIDEO STREAM

To access the RTSP video stream directly from your MFD, please refer to your MFD user manual for instructions on how to configure RTSP streams.

Real-Time Streaming Protocol (RTSP) URL links:

CAMERA	RTSP LINK
Thermal	rtsp://{IP}:8555/t1
Color	rtsp://{IP}:8555/rgb1

Depending on the model of your Watchkeeper system, one or two video streams may be available – daylight (color) and thermal.

Watchkeeper Lite provides only a daylight (color) stream, while all other Watchkeeper models include both daylight and thermal streams.

Example (Color Camera):
rtsp://172.31.201.45:8555/rgb1

Example (Thermal Camera):
rtsp://172.31.201.45:8555/t1

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9 MAINTENANCE

To ensure optimal performance, always keep your system updated with the latest available software. Regular updates provide important feature improvements, performance enhancements, and security fixes.

The camera housing and lens require periodic cleaning. Clean the lens whenever image quality decreases or visible dirt, salt, or residue builds up.

Important cleaning precautions:

- Do not wipe the lens with a dry cloth or abrasive materials such as paper or scrub brushes, as this can damage the protective coating.
- Do not use acid- or ammonia-based cleaning products.
- Do not use a pressure washer.
- Handle the windows carefully – they feature an anti-reflective coating that can be damaged by improper cleaning.

To clean the camera lens:

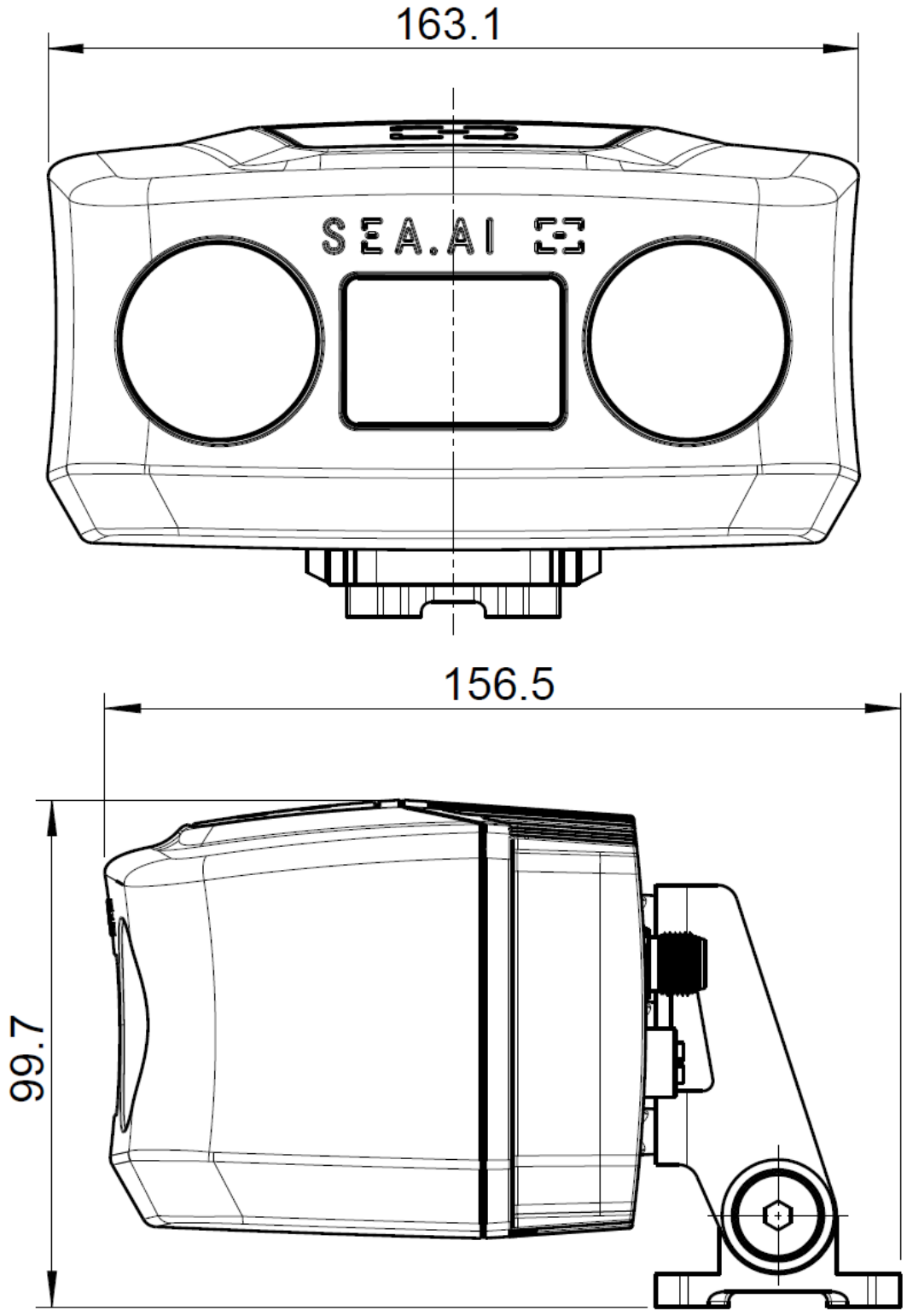
1. Rinse with fresh water to remove dirt particles and salt deposits.
2. Wipe the vision unit gently with a clean, soft cotton cloth. You may moisten the cloth with fresh water and use a mild detergent if necessary. Allow it to air dry naturally.
3. If any spots or smears remain, gently wipe the lens window with a clean microfiber or soft cotton cloth.
4. If needed, use isopropyl alcohol (IPA) or a mild detergent to remove persistent marks or residue.

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10 MECHANICAL DRAWINGS

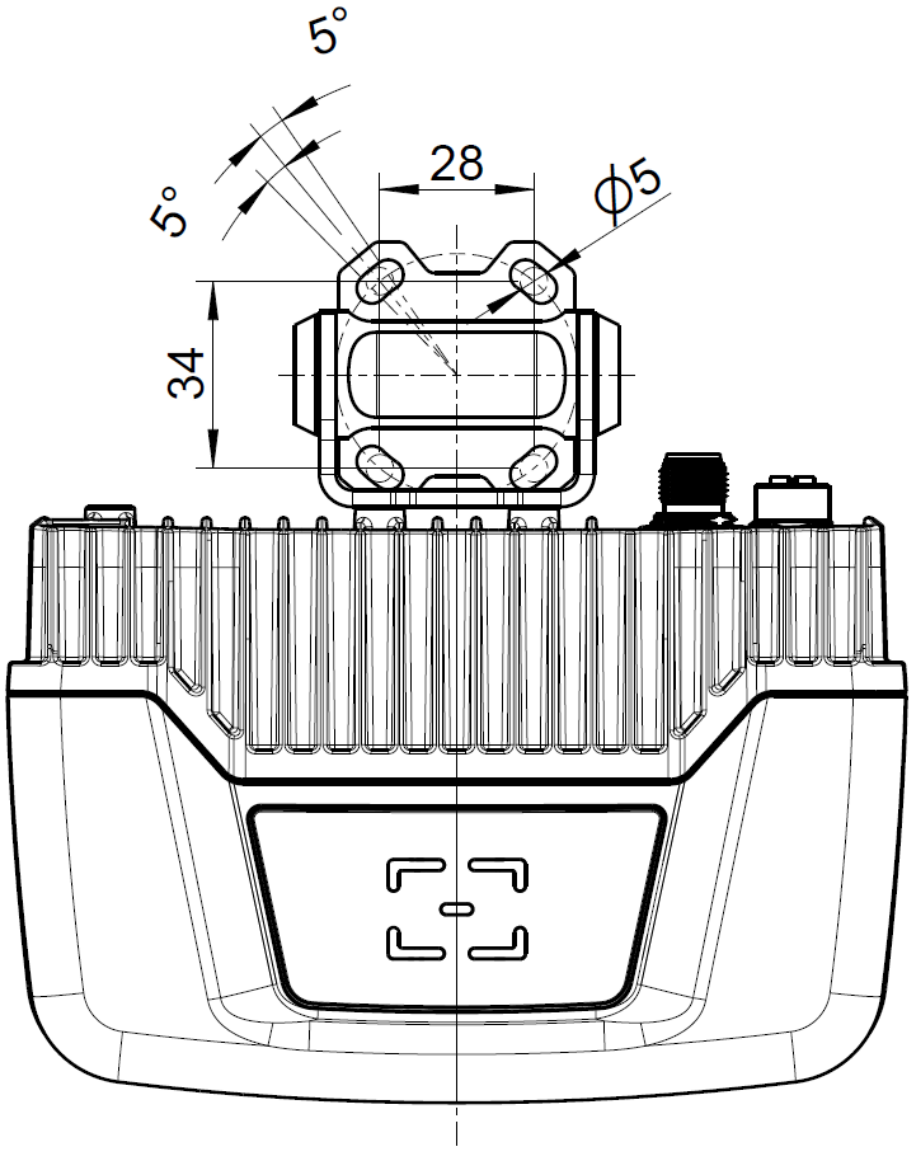
You can download Watchkeeper 2D drawing and 3D CAD files at SEA.AI Watchkeeper Document Library.

F05XXX Watchkeeper Unit (dimensions in mm)



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A04290 Watchkeeper bracket view from top (dimensions in mm)



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11 TECHNICAL SUPPORT

SEA.AI provides several options for technical support:

- Online Documentation: SEA.AI Watchkeeper Document Library.
- E-mail: service@sea.ai
- Web: www.sea.ai

The preferred method of contacting support is by e-mail. This ensures your request is properly logged, assigned, and tracked for the fastest possible resolution.

When contacting SEA.AI Support, please include the following information:

- Watchkeeper serial number
- Watchkeeper installation date
- Detailed description of the issue or problem

For worldwide support and a list of certified dealers and distributors, visit https://www.sea.ai/dealerlocator/.

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